Disc golfer's gaze control during putting: Professional versus amateur

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Spring 2025

ABSTRACT

Hirvonen, M. 2025. Disc golfers gaze control during putting, case: Professional versus amateur. Faculty of Sports and Health Sciences, University of Jyväskylä, bachelor's thesis of biomechanics, 40 pp., 2 pages of appendices.

The purpose of this study was to find, if there are differences in gaze control and quiet eye between professional and amateur level disc golfers. It also serves the purpose to increase the science basis of disc golf. Disc golf putting has not been officially researched nor the gaze control in such setting. The quiet eye can de defined as the final fixation, which is targeted on to a specific target or object, such as a disc golf basket.

The variables used in this study were the duration of quiet eye, fixation amount between the putter leaving the participant's hand and landing, gaze targeting and the success of quiet eye. These variables have been previously used in meta-analysis, therefore, they were deemed to be suitable to be used in this study as well. The three research questions were: 1. Does a professional have a better gaze control than the amateurs? 2. Does the quiet eye duration change when the distance of putting changes and are there differences between the skill levels? 3. Is there a significant correlation between the duration and success of quiet eye to the success of putting?

There were four amateurs, and four professionals involved in this study and each putted five times from four different distances (5, 10, 15 and 20 meters). For data collection purposes, the participants wore gaze tracking glasses whilst putting and quantitative methods were used to analyse the data. The results revealed that the amateurs had longer quiet eye duration than professionals, although both skill levels had an average total from all the putts of over 2000 ms. On most participants the quiet eye duration decreased when the distance increased. This was mostly with both skill levels. There were little differences between the skill levels with the amount of fixations. However, a fewer amount of fixations had a connection of resulting to a better putt result. Furthermore, there were no significant differences in the gaze targeting between the two skill levels and significance only on one distance on the success of putting results. The success of quiet eye had significance from the 15-meter distance. Finally, the results showed that the quiet eye duration had no significant correlation to the success of putting.

According to the results, professional and amateur disc golfers have some, but fairly little differences in gaze control and quiet eye duration. What was more significant was the amount of fixations between the throw and landing of the putter and the success of quiet eye. Based on these results, it can be recommended that disc golfers and coaches give gaze controlling and targeting more emphasis in putting training, however, more research on the matter needs to be done.

Keywords: Disc golf, Quiet Eye, Gaze Control, Skill level

TIIVISTELMÄ

Hirvonen, M. 2025. Frisbeegolfaajan katseenkäyttäytyminen puttaamisen aikana, tapaus: Ammattilainen versus amatööri. Liikuntatieteellinen tiedekunta, Jyväskylän yliopisto, liikuntabiomekaniikan kandidaatin tutkielma, 40 s., 2 sivua liitteitä.

Tämän tutkimuksen tarkoituksena oli selvittää, onko ammattilaisen ja amatöörin välillä eroja katseen hallinnassa sekä quiet eye ilmiössä puttaamisen aikana. Työn tarkoituksena oli myös tuottaa lisätietoa frisbeegolfista sekä tieteestä sen taustalla. Aiempaa vertaisarvioitua tutkimustietoa ei frisbeegolfin puttaamisesta tai katseen käyttäytymisestä ja quiet eye ilmiöstä ole olemassa. Quiet eye voidaan määritellä katseen viimeisenä fiksaationa, joka on kohdennettu ennalta määritettyyn kohteeseen, kuten frisbeegolf koriin.

Tutkimuksen muuttujina käytettiin quiet eye -kestoa, fiksaatioiden määrää putterin kädestä irtoamisen ja sen laskeutumisen ajan välillä, katseen kohdentamista sekä quiet eye onnistumista. Useaa näistä muuttujista on hyödynnetty aiheeseen liittyvissä meta-analyyseissä, joten näiden muuttujien voitiin olettaa sopivan myös tähän työhön. Tutkimuksen kolme tutkimuskysymystä olivat: 1. Onko ammattilaisella parempi katseen hallinta puttaamisen aikana kuin amatöörillä? 2. Muuttuuko quiet eye -kesto puttausmatkan muuttuessa ja onko taitotasojen välillä eroja? 3. Onko quiet eye -kestolla ja sen onnistumisella tilastollisesti merkitsevää yhteyttä puttaamisen onnistumisessa?

Tutkimuksessa oli mukana neljä amatööriä sekä neljä ammattilaista. Heistä jokainen puttasi viisi kertaa neljältä eri heittopisteeltä (5, 10, 15 ja 20 metriä). Tutkittavat käyttivät katseenseurantalaseja tutkimuksen aikana, minkä avulla data kerättiin. Tutkimuksessa käytettiin kvantitatiivisia tutkimusmenetelmiä. Tulosten mukaan amatööreillä oli keskimäärin pidempi quiet eye -kesto kuin ammattilaisilla, vaikkakin molemmilla tasoryhmillä keskiarvo kaikista puteista oli yli 2000 ms. Useimmilla tutkittavilla molemmissa tasoryhmissä quiet eye -kesto lyheni matkan kasvaessa. Fiksaatioiden määrissä ei ollut suuria eroja tasoryhmien välillä, mutta vähäisempi fiksaatioiden määrä oli yhteydessä parempaan puttaustulokseen. Katseenkohdistamisessa ei myöskään ollut tilastollisesti merkittäviä eroja tasojen välillä. Tämän lisäksi vain yhdellä puttausmatkalla oli tilastollinen merkitsevyys puttaamisen tuloksen ja katseen kohdistamisen välillä. Lopulta myöskään quiet eye -kestolla ei ollut tilastollista merkitsevyyttä puttien onnistumiseen.

Tulosten mukaan ammattilaisten ja amatöörien välillä ei ole suurta eroa katseen hallinnassa ja quiet eye -kestossa. Merkittävää oli kuitenkin fiksaatioiden määrän sekä quiet eye -onnistumisen yhteydellä puttaamisen onnistumiseen. Näiden tulosten perusteella voidaan suositella, että frisbeegolfaajat sekä valmentajat antavat lisää painoarvoa katseen hallinnan kehittämiseen harjoittelun yhteydessä. Toisaalta lisää tutkimusta aiheesta vaaditaan.

Avainsanat: Frisbeegolf, Quiet Eye, Katseenhallinta, Tasoryhmä

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1 INTRODUCTION

Many different sports, such as darts, golf, billiard and football, include targeting tasks. These targeting tasks include gaze control and the quiet eye phenomenon. These skills can be trained and therefore are often more developed in professional athletes compared to novice athletes. The quiet eye is defined as the final gaze fixation on a specific target that lasts at least for 100 ms and stays in a visual range of 3 degrees. (Vickers 2007, 11) Professional athletes have been found to be able to process information more rapidly and thoroughly from gaze targets than athletes of lower skill level. Furthermore, professionals have been found to have fewer fixations with longer durations in targeting tasks, which often leads to better results. (Mann et al. 2007)

Disc golf is a relatively new sport when it comes to professionalism. However, during the previous decade the sport has taken massive steps and is developing its professionalism continuously by organizing more events, having more competitiveness and handing out more price money (PDGA 2025). Disc golf is a throwing sport, which includes maximal power, precision shooting and targeting tasks. These characteristics require the players to have explosive strength and mobility (Tervonen 2022).

The purpose of this study was to bring more knowledge of both, disc golf and gaze control. It also serves the purpose to showcase the possible differences that amateurs and professionals have between each other in disc golf gaze control while putting. The previous research on disc golf is scarce, so the original purpose of this study was to increase the research basis and professionalism of disc golf.

The literature review consists of previous research on gaze controlling, quiet eye, biomechanics of disc golf and the previous research and basics of disc golf. The study uses quantitative methods in data collection and analysis.

2 DISC GOLF

Disc golf is a sport that has a low threshold of starting when it comes to costs and skills. The costs are low, since new discs are very affordable and being provided in many different stores both online and on-the-spot. Furthermore, most of the courses are free to play (PDGA 2024). The building of new courses is seen to be one of the most cost-efficient choices for cities in terms of recreational outdoor activity for citizens (Anttila 2016). The courses vary from easy to elite level and the sport can be played from children to elderly people. However, during recent years the sport has grown rapidly and has also become more competitive and professional. (PDGA 2024)

2.1 Basics of the sport

The rules of disc golf are similar to (ball) golf. The main aim is to get the disc into a basket with as few throws as possible. Usually, a disc golf course consists of either 18 different holes or 9 holes which would be played twice, as a full round of disc golf means 18 holes. The hole starts from a designated tee pad in which the first throw takes place. After that, the player continues to throw the disc from the spot it landed, unless the disc is thrown out of bounds. The hole ends in a metal basket, which can vary slightly for example in height and the type of chains it has. (PDGA 2024)

There are different types of discs to use whilst playing disc golf. In fact, it is one of the most variable sports in terms of holding the most variety in sports equipment projectile shapes (Giljarhus et al 2022). The four most common types of discs are distance drivers, fairway drivers, mid-ranges and putters. Distance drivers are mainly used to gain the maximum distance. Fairway drivers are more situational discs for example used for curvature or to gain a long distance with more control. Mid-ranges are used for control and approach shots and putters for shorter approaches and putting the disc in the basket.

There is no one right way to throw a disc in disc golf and many players have a variety of techniques for different situations. However, the main three types of throws are backhand, forehand and putting. Backhand is the most common way of throwing (Bocks 2019) and is the way to also gain the maximum amount of distance among the elite players when thrown correctly. Backhand throw has a unique technique, since there is not a similar throwing

technique in other sports (Holcomb 2023). Forehand throw somewhat mimics the technique of a baseball throw and putting again is totally unique to any other throwing technique.

For a player to be competitive in the sport and to be able to participate in the official competitions, they must register themselves as an official member of the Professional Disc Golf Association (PDGA). (PDGA 2024) Once a player competes in official PDGA competitions they receive a rating depending on their performance. The rating is based on the rating of the course played on, also known as Scratch Scoring Average (SSA). The player's result on the course is compared to the course rating, which leads for the player to receive a rating of their own. The higher the player's rating is, the better they have performed. Receiving a rating of 1000 or more means that the player has played a lower score than the SSA is on the courses. The top-rated player, Gannon Buhr, in November 2024 held a rating of 1054. (PDGA 2024)

There are different tiered competitions a player can participate in. These competitions have different requirements for example in membership status and level of player. The lowest is tier-C, which do not require a PDGA membership. B- and A-tier require an active PDGA membership. The highest ranked competitions are the disc golf pro tour and the PDGA majors, which include, for example, the World Championships. The higher ranked the competition is the bigger the win payouts are. However, to be able to register for the highest two rated competitions, players in MPO (mixed professional open) category need to have a rating of at least 935 and in FPO (women's professional open) a rating of at least 825. (PDGA 2024)

2.2 Key figures of the sport

The sport of disc golf was created in the 1970s in California, the United States of America. The very first player registered themselves in 1976 as a member of the PDGA. Figure 1 shows the growth of the sport in terms of official memberships amounts. The initial growth was slow until the late 2010's. The 100000th member milestone was reached in 2017 – almost 40 years after the first player (Innova 2022). However, already in 2021, the membership amount had doubled to 200000. The growth has not stopped since, and in the end of 2023, the total registered membership amount was over 250000. (PDGA 2024) It is important to note that these figures do not reflect the number of active memberships now, but more the total amount of PDGA

memberships issued, since 1976. The number of active memberships was 136636 globally and the newest issued PDGA membership number was 275649 (PDGA 2024)

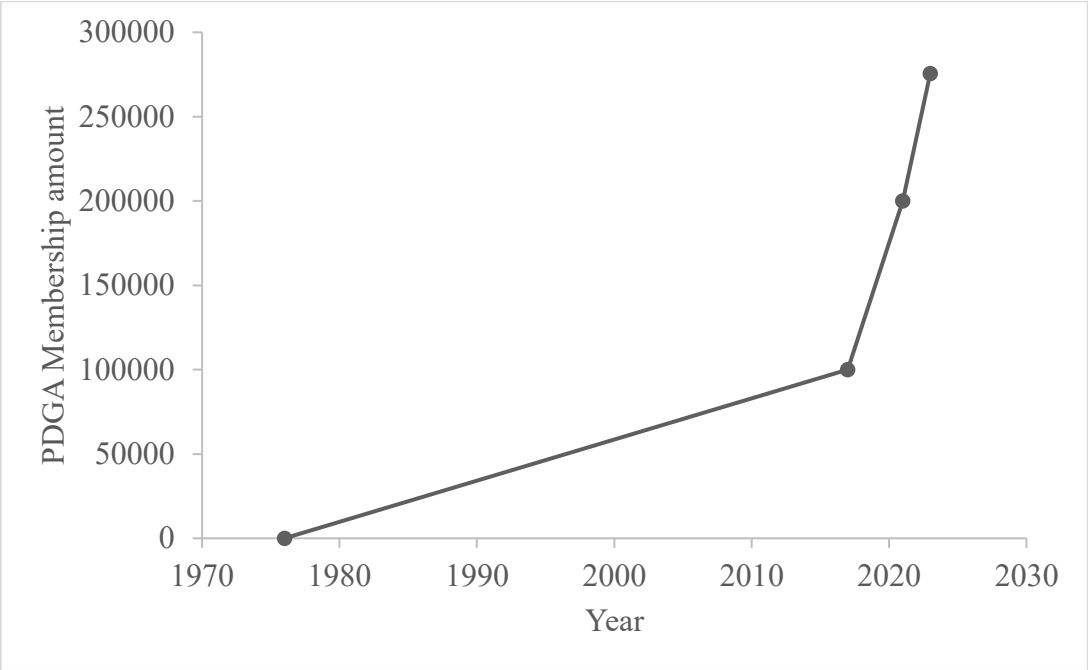

FIGURE 1. The growth of official registered PDGA memberships since 1976, figure created based on PDGA 2024 data. (PDGA 2024)

Although disc golf was founded in the United States, it has become a global sport. Figure 2 shows the countries to have at least one active PDGA membership. Furthermore, all but 12 countries, had more than one active membership. (PDGA 2024). As the figure 2 shows, disc golf is a global sport. According to data from 2023, there were 15205 disc golf courses globally. These are in over 85 different countries, from which Finland has the most per capita - 1068. It was evaluated that approximately 4 courses were built each day during 2023 globally. (Udisc 2024)

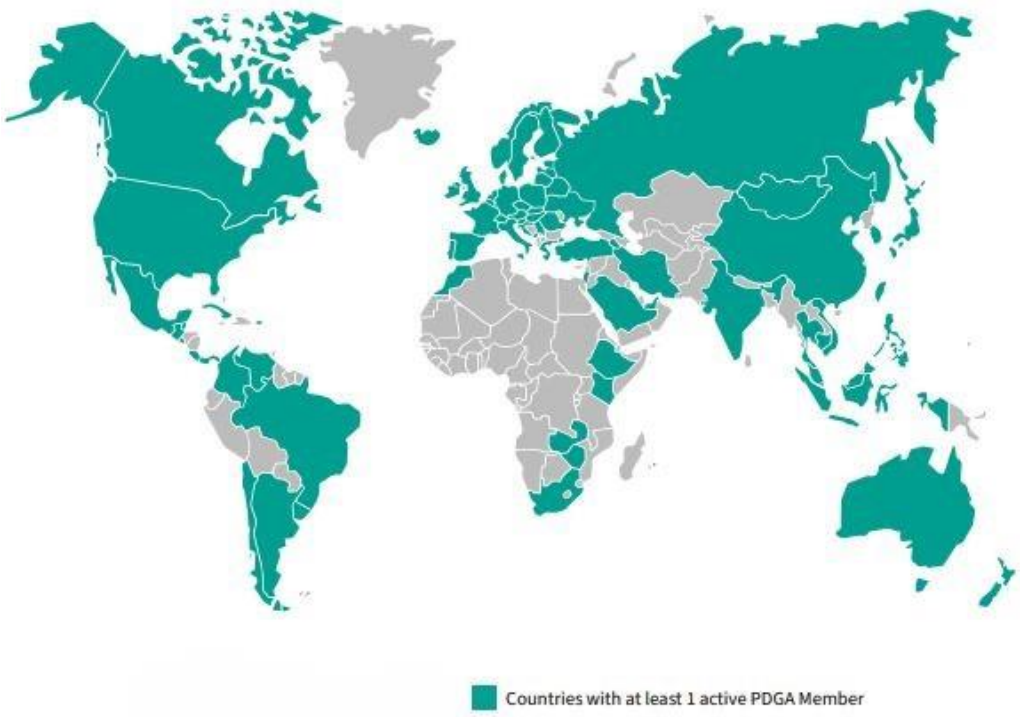

FIGURE 2. Countries to have at least one active PDGA member by the end of 2023. (PDGA 2024, altered)

It is important to note that the number of official memberships does not cover all the players of the sport, these figures rather tell more of the professional and competitive growth of the sport. The actual number of players in disc golf, globally, exceeds the PDGA membership player amount widely. It has been evaluated that there are more than 4000000 people globally who play disc golf (Innova 2022). According to an official survey and research, there are approximately 1000000 people in Finland who play disc golf, which supports the previous global evaluation (Finnish Disc Golf Association 2024).

2.3 Previous research on disc golf and main findings

The amount of scientific research on the sport of disc golf is scarce. For example, when searching in October 2024, for peer reviewed articles using only keywords “disc golf” and/or “frisbee golf” on the data bases PubMed and SportDiscus the number of results was 26 on the former and 20 on the latter. Furthermore, in the PubMed’s results there were also some (ball)golf results shown together with disc golf. To highlight the scarcity of research done on disc golf, when PubMed was searched with the keyword “golf”, more than 3800 results were

found. Finally, when these data bases were searched with the keyword “frisbee” SportDiscus showed more than 100 results and PubMed more than 300. In SportDiscus more than half of these results were related to ultimate frisbee and on PubMed approximately 12% were related to ultimate frisbee. It is important to understand that disc golf and ultimate frisbee are very different sports between each other. The biomechanics of the throws are different (Leppilahti 2019), and the discs used are different. Therefore, the research done on ultimate frisbee is rarely applicable on disc golf in a biomechanical sense.

In the time of writing this thesis, the backhand throw was mainly the only throwing technique that has been scientifically researched is disc golf. There is one study about the putting in disc golf and one about the forehand throw, but with ultimate frisbees. Other studies have researched injury prevention, the social phenomenon of disc golf, environmental impacts, physical education and aerodynamics of the discs. Because of the nature of this thesis, mainly the biomechanical, physiological and aerodynamical papers will be covered.

Hummel’s (2003) research is one of the first ones to cover the analysis of frisbee flight and biomechanics of backhand throws. However, it is important to highlight that Hummel’s thesis researched frisbees and not disc golf discs. Also, the throw technique analysed was one used in ultimate frisbee, in which the technique differs from the disc golf backhand throw. Nonetheless, Hummel’s 76-page thesis is an asset for any frisbee or disc related research and provides valuable knowledge. The thesis is split into two separate projects, the frisbee flight characteristics and the biomechanics of frisbee backhand throw.

Leppilahti (2019) conducted the first kinematic analysis of the backhand throw, in which the backhand throwing technique was analysed with a 3D movement analysis. The purpose of the study was to make findings of which kinematic factors impact the most on the release speed of the disc as well as to bring more knowledge of the backhand throw, from a biomechanical aspect.

3 BIOMECHANICS OF DISC GOLF

The biomechanics of disc golf has not been studied as thoroughly as many other throwing sports, such as the javelin throw, darts or a baseball throw. This can probably be both because of the sport being relatively new, especially professionally, and for having various throwing techniques within the sport. However, disc golf does follow the same biomechanical laws as other throwing sports – depending on the type of throw that is used. A purpose of the throw is to displace an object with a range of motion and usually to grow the objects momentum using several body segments (Enoka 2015, 155).

3.1 Kinematics and the proximal distal order

Kinematics is the science of motion, but it does not research the causes of the motion such as forces. The three factors behind the motion are velocity, position and acceleration. (Enoka 2015, 3) Kinematics can therefore be used to explain the phenomenon of throwing. Most often the throwing movement includes three-dimensional kinematics (Enoka 2015, 155).

When optimal throwing motion is the goal, the motion usually follows in a segmental order from the proximal to the distal end (PD), which results to higher velocity in the distal parts of the body, because of the proximal parts' angular velocity (Putnam 1993). In many throwing sports, the throwing motion follows the PD order, which usually starts from the thrust of the lower body limbs, then followed by the pelvic area rotation as well as the trunk and finalised with the movement of the throwing arm (Serrien & Bayens 2018). This is true in many throwing sports such as in tennis serving, volleyball spiking and team-handball throwing. As stated, the PD order follows in the order from proximal to distal parts of the body through joints, which is also shown in figure 3. (Wagner et al. 2014).

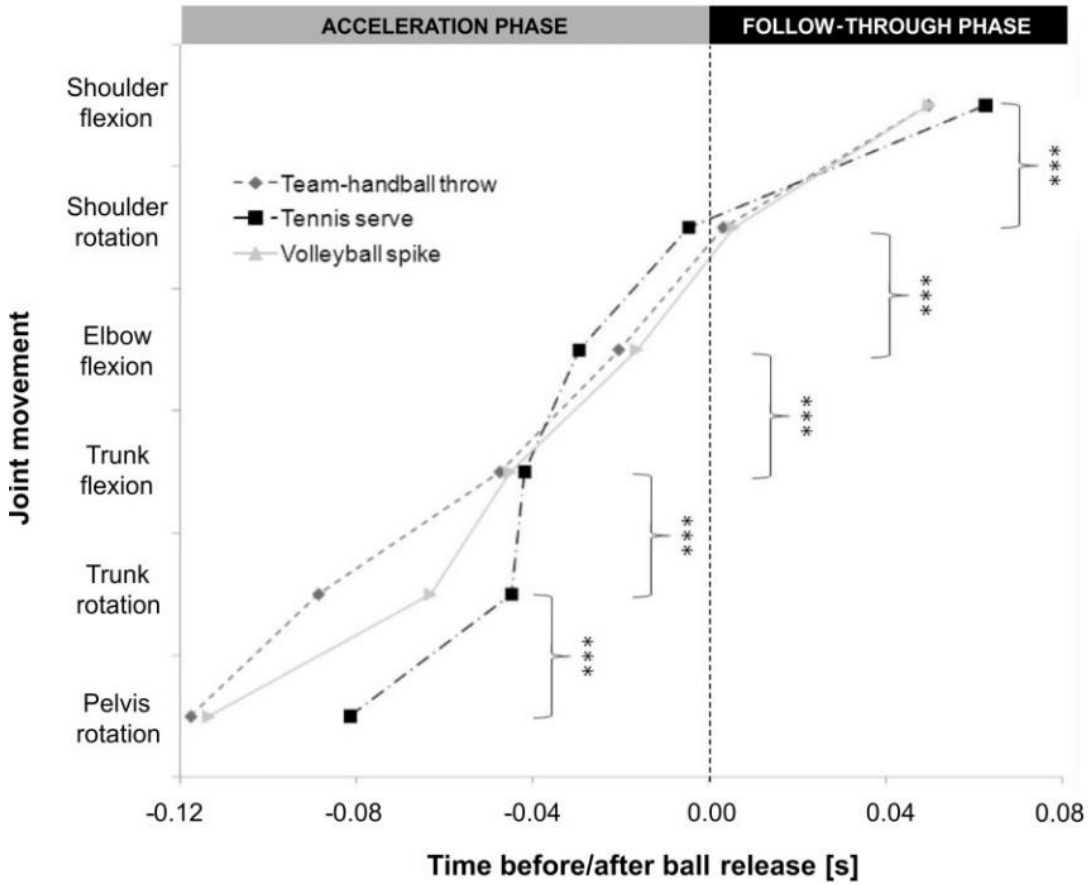

FIGURE 3. The mean timing of maximal angular velocities in the team-handball throw, tennis serve, and volleyball spike. Significant differences between two joints: *P < .05, **P < .01, ***P < .001. (Wagner et al. 2014)

The kinetic chain or kinematic chain is a concept used in biomechanics and is originally derived from the science of engineering. The kinetic chain in humans consists of rigid segments connected with joints. Once a force is applied to the distal part of the chain the motion follows through as a chain reaction, generating positive force. (Karandikar & Vargas 2011). According to Karandikar & Vargas (2011), in his book *Kinesiology of the human body* (1955), Steindler was the first to define the kinetic chain into two different motions, the open kinetic chain (OKC) and the closed kinetic chain (CKC). In an OKC motion the terminal end of the chain is the only one to move freely, when force is applied. However, in the CKC motion, the entire chain moves when force is applied to the terminal end of the chain. Most motions in sports are CKC motions, especially when the initial activity begins from a steady stance and the force transferred from the motion follows through the chain to more mobile segments. (Karandikar & Vargas 2011)

3.2 Distance throwing

The very core of disc golf is to get the disc into the basket with as few shots as possible (PDGA 2024). There are also only a few restrictions of how a player can throw the disc and the restrictions mostly concern the position from which the throw takes place (PDGA 2024) and not so to the way the disc is thrown. However, the shape of the disc cannot be altered, for example cut or squeezed as a ball (PDGA 2024). This is mainly because dents, stretching of the plastic or other alterations impact on the flight of the disc (Giljarhus et al. 2022). Therefore, players have come up with many different throwing techniques in disc golf (Nelson et al. 2015). The choice for a specific throwing technique can be caused by many different reasons. These are for example: The shape of the needed throw, weather aspects such as wind, blocking objects and distance to the basket (Nelson et al. 2015). Also, players vary in skill and anthropometric measurements, which both affect in throwing the disc, therefore different throws might suit better for different players. The most used throwing technique in disc golf, however, is the backhand throw (Bocks 2019).

Hummel (2003) researched the biomechanics of a submaximal stand still backhand throw of an ultimate frisbee. One of the main findings was that the throw did not entirely follow the PD order. The tests showed that all parts of the kinetic chain, but for one, generated positive force. The elbow flexion was found out to generate negative force throughout the throwing motion, which goes against the kinetic chain principal. (Hummel 2003)

In Leppilahti's (2019) maximal backhand throw research, however, the results were different. One of the main findings was that the backhand throw of a disc golf disc mainly follows the PD order from the thorax to the wrist. The forces generated were all positive and the angular velocity of the joints grew throughout the throw the more distal the joint was. The only part that did not follow the PD order was the external rotation of the shoulder, because with six of the participants the angular velocity grew steadily even after the follow-through of the throw. However, with two participants the throw followed the PD order completely. Leppilahti (2019) argued that the main reasons for the differences in the results compared to Hummel (2003) were that the technique of the throws are different because the latter was a submaximal stand still throw whereas Leppilahti's (2019) test involved a maximal throw with a run up. (Leppilahti 2019)

3.3 The biomechanics of putting

When throwing an accurate throw, such as darts throwing or disc golf putting, one needs to keep the throw as composed as possible (Enoka 2015, 157) The research behind putting in disc golf is almost non-existent, therefore not much can be said about it. Jackson and Tanner (1993) conducted an analysis about elite and novice disc golfers' putting. However, it is important to mention that this work has not been peer reviewed and is, therefore, not entirely reliable data. They compared two elite and two novice players against each other in putting from 10 meters. Their movement in horizontal, sagittal and frontal plane were analysed with a highspeed camera. Their main findings were that the elite players were more composed in the putting and had more simultaneous intersegmental coordination than that of the novice players. (Jackson & Tanner 1993)

When the distance of the throw or putt changes, the laws of force change as well. For example, in a baseball long-toss throw, if the throwing distance grows, the arm speed and shoulder external rotation increases, whilst the arm slot angle decreases (Dowling et al. 2018). Although, this has not yet been researched, it can be assumed that disc golf putting holds different changes in throwing forces when the putting distances grow.

3.4 Basic aerodynamics and gyroscope of the discs

There are multiple forces that impact the flight of the disc, which can be observed in figure 4. Drag (**D**) is one of the main factors to slow down the disc and it acts as an opposite and parallel force to the velocity. The magnitude of the drag can change during the flight of the disc but can never be zero. The other force to slow down the disc is gravity (m**g**), which is a constant force and acts at the centre of mass trying to pull the disc down on the ground. (Hummel 2003)

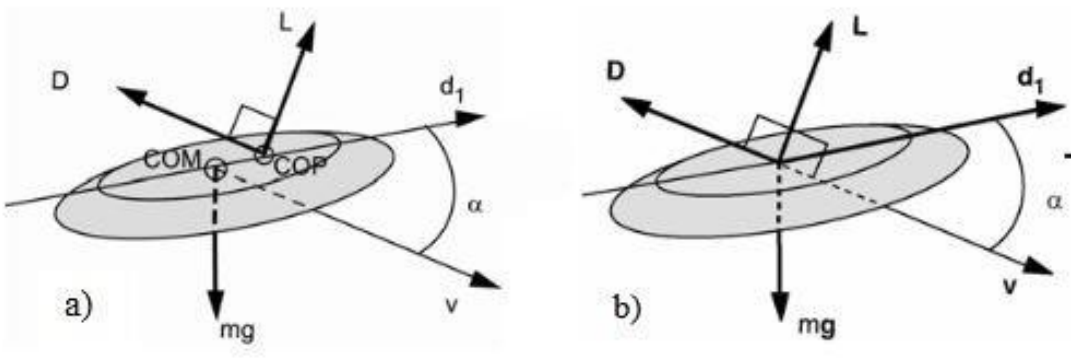

FIGURE 4. Forces acting the flight of the disc. **D** (drag) and **v** (velocity) act as opposite vectors and **L** (lift) acts perpendicular to these. The angle between the velocity vector and the direction vector (**d**$_1$) is known as the angle of attack (α). Gravity (m**g**) is also a force that impacts the flight of the disc. (Hummel 2003)

When the disc is flung from the hand, it meets air flow. This air flow is split from the nose of the disc and, if the angle of attack (α) is zero or more, the curvature of the disc causes the air flow to go beneath the disc. As the air flows beneath the disc, the disc pushes the air down and by Newton's third law, the air generates an equal opposite pushing force to the disc, called lift. Lift acts on the disc as an opposing force to gravity, therefore, lifts the disc away from the ground. However, it does not act directly to the opposite direction compared to gravity, but the direction depends on the angle of attack. The magnitude of lift is also dependant on the angle of attack of the disc. (Hummel 2003)

It is important to understand that the centre of mass (CoM) and the centre of pressure (CoP) are not at the same point on disc. The CoP is the point where drag and lift acts on the disc and it can vary during the flight of the disc. The angle between the plane of the disc, also known as the nose of the disc, and the velocity vector is known as the angle of attack (α) as can be seen from figure 4. The degree of α has an impact on the magnitudes of both lift and drag. (Hummel 2003)

As the α affect the drag and lift, it also affects the pitching momentum, i.e. the momentum for the disc to roll during the flight. According to Kamaruddin et al. (2018) in the aim of reaching maximum distance with a disc, one must maximize the lift-to-drag ratio and minimize the pitching momentum with any given α. According to multiple research the α should be positive for better distance gain, but too high of an α will also slow down the disc or cause it to pitch

(Hummel 2003; Kamaruddin et al. 2018; Giljarhus et al. 2022). For example, Hummel (2003) suggested that an angle of attack of nine degrees could be optimal and in Kamaruddin et al. (2018) research the optimal α, in terms of maximal lift-to-drag ratio and minimal pitching momentum, was somewhere between 5-8 degrees, depending on the type of disc.

When the disc is thrown it acts as a gyroscope and revolves around its axis as is shown in figure 5. As the disc spins, it faces aerodynamical forces and throws, where especially accuracy and control are sought, gyroscopic stability is key for the disc to resist these aerodynamical forces. The spin affects the disc to be more stable in its flight, because of angular momentum. The higher the angular velocity of the throw is, the higher the angular momentum is. This again leads to the disc being more stable, because it conserves angular momentum. (Koch et al. 2024)

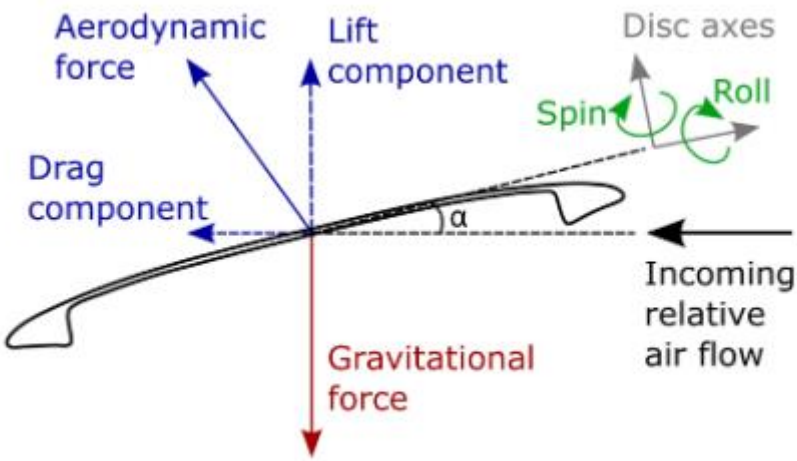

FIGURE 5. Momentums and forces acting on the disc. (Giljarhus et al. 2022)

4 QUIET EYE

Different parameters have been researched when it comes to elite sports and performance. Cognitive skills such as visual processing, execution of motor skills and decision making are key skills for athletes (Klostermann et al. 2018). The gaze control and quiet eye, which are perceptual-cognitive skills, have been widely researched (Mann et al. 2007). However, these skills have not yet been researched in the field of disc golf.

4.1 The basics of visual system

When the human receives light stimulation through the pupil into the eye it becomes something to be seen. The pupil's size adjusts according to the amount of light available and is then bent through the cornea and the lens. The bent light is then processed in the fovea, which is responsible for visual clarity. The person must align their sight with the fovea to see objects and targets clearly. Because of the size of the fovea, the total area, which can be seen clearly is small, only 2-3° of the visual angle. (Vickers 2007, 18)

The vision field of the human is split into two, left and right, as is described in figure 6. When the light reaches the fovea, visual receptors, retinal ganglion cells send action potential forward in the visual system for the brain to process the information. The action potential moves along the optic tract and crosses over at the optic chiasma. The action potential then reaches the pretectal nucleus, which manages the reflexes of pupil caused by light, and then the lateral geniculate nucleus (LGN). There, the visual information is forwarded to different optic radiations, which then process information brought from different areas of vision, the lower and upper areas as well as from the focused vision area. The superior colliculus adjusts the movement of the eyes. From the LGN the information goes to the primary visual cortex, where the vision information is finally processed to knowledge of what and where we see the object. (Risto 2023, 151)

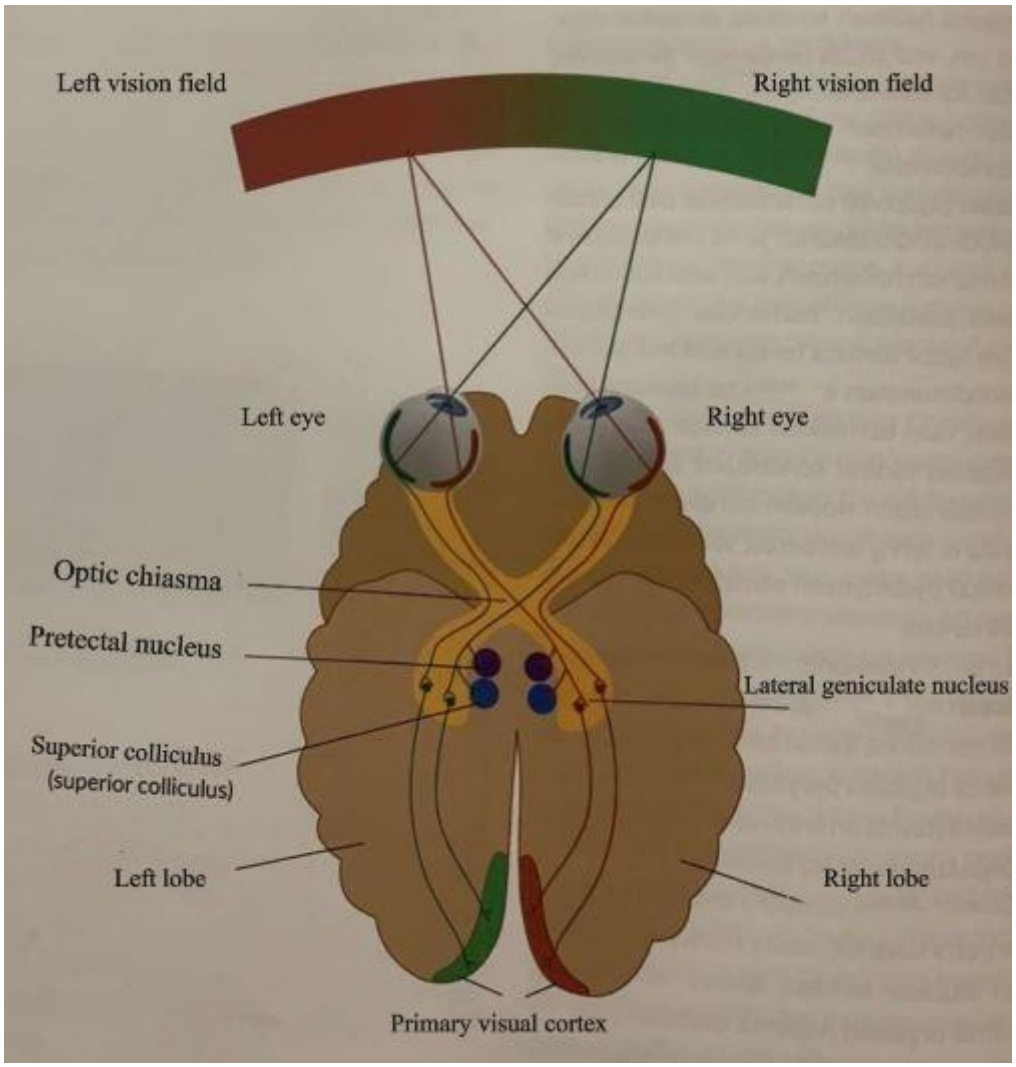

FIGURE 6. The visual system in the brain. (Risto 2023, 150 altered)

4.2 Targeting tasks and gaze control

When a person is completing a task, it requires skills and understanding. Skills can be roughly divided into open and closed skills. The more unpredictable the environment is, where the skill takes place, the more open it becomes. Closed skills occur in a more predictable environment. (Schmidt et al. 2018, 25-26) An example of the latter is darts throwing, where the environment is very stable and predictable. An open skill is for example playing football, where the environment is constantly changing. Golf putting can be defined more of a closed, self-paced skill. This is because, although, for example, the weather can change during the putting event, the changes in the environment are semi-predictable, hence the skill is more closed (Schmidt et al. 2018, 26).

There are two types of gaze control: One that is focused on an object or a target for a notable set of time and can be processed by the brain. The other is unconscious, because of the speed of the gaze and, therefore, cannot be processed consciously. (Vickers 2007, 19) Furthermore, the gaze control does not work similarly with each sporting task but can be categorised to three different categories. These are tactical, interceptive and targeting tasks. The gaze control differs with each of these tasks. (Vickers 2007, 10)

When a person is completing a targeting sporting task they need to focus on a set object. Furthermore, the main goal often is to be accurate and consistent (Vickers 2007, 68). First, the person needs to identify the object where to focus on. These can be internal, external-proximal and external-distal focus points. (Chen et al. 2022) An example of internal focus can be the throwers own arm, the external-proximal for example the disc and the target basket is an example of the external-distal focus of attention. Furthermore, the gaze control in targeting tasks can be divided into three different tasks: gaze control to moving, abstract or fixed targets. A fixed target is something that is stable for example a basketball hoop. For a moving target the target of aim and gaze is in motion, such as in team-handball throw. An abstract target for gaze and aim is something that is fixed in space but has an abstract nature for example a golf hole on a sloped green. (Vickers 2007, 68)

4.3 The quiet eye and main research findings in different sports

The term quiet eye (QE) was first used by Vickers (1996) in a paper about basketball free throwing. The term is defined as the final fixation of the eye, that is located to a specific target. For the fixation to meet its requirements, it needs to last at least for 100 ms and be within three degrees or less of the visual angle. Once completing a task, for example a golf putt, the onset of the QE happens before the last movement of the said task. The offset occurs when the gaze deviates for more than 3° of the visual angle at least for a duration of 100 ms. (Vickers 2016) Plenty of research has been done on the QE in different sports. One of the main findings have been, that athletes with better success in tasks, such as throws and putts, have recorded a higher QE duration than that of the worse performers or results.

Xu et al. (2021) used electromyography (EMG), electrooculography (EOG) and electroencephalography (EEG) to record QE data from golf putting. They discovered that successful golf putts were related to longer QE durations than the failed putts. They recorded a total QE duration difference of more than 100 ms between hit and missed putts. Furthermore, it was discovered that the longer QE duration was linked with better motor performance.

Walters-Symons et al. (2018) conducted research, in which they sought to find out, if QE duration changes when a task is made more difficult. In this experiment they increased the difficulty of golf putting by manipulating different elements. Their findings were that more complex tasks resulted in a longer QE duration. However, they pointed out that this area needs further research as some results of the research were unclear.

Giancamilli et al. (2022) conducted research, in which they analysed how the QE changes when time and performance pressure is included in the basketball free throws. A total of 34 participants were included in the results of the study from, which 18 were ranked as competitive-elite and 16 semi-elite. The main findings were that competitive elites recorded longer and earlier QE duration in trials with no time pressure added. Time pressure decreased both QE duration as well as free throw accuracy. The QE duration and free throws made were also higher amongst competitive elites when performance pressure was added, but no time pressure was included.

Klostermann and Moeinirad (2020) conducted a meta-analysis of gaze behaviour. A total of 101 studies were included in the meta-analysis, which covered different gaze behavioural elements between experts and intermediates as well as experts and novices. One of the main findings were that more studies support the hypothesis that experts have a longer QE duration than lower skilled players as can be seen from figure 7. The number of studies supporting the theory was 17, and against it there were 4 studies. It was pointed out, however, that many of said studies had a similar research design and no further differentiation could be done.

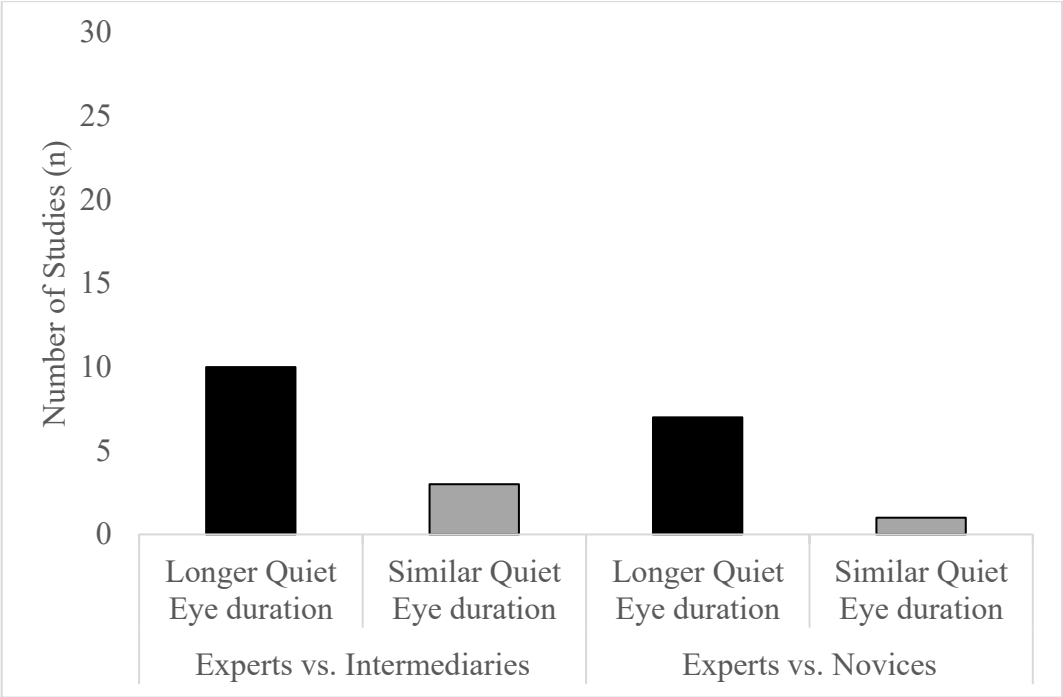

FIGURE 7. Number of studies comparing the QE duration between experts and intermediates and experts versus novices. (Klostermann & Moeinirad 2020, altered)

Reunanen (2024) conducted the most recent gaze behavioural study. The purpose of this research was to find out if there are gaze behavioural differences between expert and intermediate javelin throwers whilst throwing a javelin from a run-up. A total of 11 athletes were researched and they were divided into experts and intermediates based on their throwing records. The main findings were that the experts (408 ms) had significantly longer QE durations than the intermediate (180 ms) throwers. This means that experts had over twice the duration of the QE.

5 RESEARCH QUESTIONS AND HYPOTHESIS

Research questions 1: Does a professional disc golfer have a better gaze control than an amateur in a putting environment?

Hypothesis 1: It can be assumed that the professional disc golfer has a better gaze control than the amateur

Justification for the hypothesis 1: This can be, because perceptual-cognitive skills are often more developed with athletes who play on a professional level and the gaze control can be trained (Vickers 2007, 11).

Research questions 2: Does the quiet eye duration change when the putting distance changes and are there differences between the two skill levels?

Hypothesis 2: It can be assumed that the amateurs have more change in quiet eye duration than the professionals. Furthermore, it can be assumed that the professionals can maintain or even lengthen their QE duration.

Justification for the hypothesis 2: For amateurs it can be assumed that the quiet eye duration changes the most, because they have less experience and certainty of making longer putts, hence the gaze might wonder around. Previous research shows that for professional athletes, the QE duration increases, when performance difficulty increases. (Walters-Symons et al. 2018; Giancamilli et al. 2022)

Research questions 3: Does the duration and success of quiet eye have a correlation with the success of putting?

Hypothesis 3: This can be assumed to be true.

Justification for the hypothesis 3: Previous research has proven that with a longer QE duration a better performance can be achieved, especially in targeting tasks (Xu et al. 2021). In many sports, such as in basketball, the better the free throw success is, the longer the quiet eye duration has been found to be (Giancamilli et al. 2022).

6 RESEARCH METHODS

As no previous similar disc golf research had been done, this study took advice from similar gaze control studies. However, the main methodology ideas were conductors own. The research methods used in this study were quantitative. All the participants chosen were contacted via phone, social media or email.

6.1 Research participants

The participants were chosen based on multiple criteria. For a participant to be chosen, they needed to be at least 18 years old, male, right-handed and either a professional on disc golf or a long-term amateur. Furthermore, to be eligible to be a test participant, they could not have or wear prescribed glasses nor contact lenses of their own, even though the manufacturer has stated that contact lenses do not affect the results of gaze estimation (Tonsen et al. 2020). This was because they would wear special gaze measuring glasses throughout the test, which despite the manufacturers research of not having an impact, was feared of not working properly should the participants wear glasses or contact lenses of their own. For a player to be classified as a professional disc golf player, they needed to have their current player status as "professional" on the official player registration website. Furthermore, the professionals needed to have at least a current rating of 1005. For a participant to be classified as an amateur they needed to have at least two years of active disc golf playing history behind and not have a professional status nor have participated in competitions above MA2, which means they could not have a rating of over 935. In total of four professionals and four amateurs were chosen as participants.

All participants filled in a query, in which they confirmed to be healthy and gave their acceptance to be participants. Furthermore, their age, gender, height and weight were asked. The participants were then told their rights as a participant, meaning they could quit the test at any time, if they wanted to. They were then told thoroughly what they would be doing during the test.

6.2 Research design

All the participants completed the same preplanned warm-up, which consisted of jogging for three minutes, dynamic movements and short full body stretches. A warm-up programme created by a disc golf professional was used as a vague guide (Laine 2017). After that they were allowed to do practice putting for approximately five minutes from any distance they wanted to. The purpose of practicing and warm-up was to ensure the readiness of the participants from the very first throw. Before the test began, they were allowed to do a few practice putts with the glasses on. Finally, before the actual test, the glasses were calibrated per participant for the gaze to be accurate and data valid, although the manufacturer has stated that a calibration is not always necessary (Tonsen et al. 2020).

The tests were completed either in the gymnasium of the University of Jyväskylä Sports Science facilities or in a different gymnasium, which was more suitable for a participant location-wise. All the tests were completed indoors and on a similar synthetic surface. The test was conducted by the director of the research and there was also a scribe to mark down the scores of the throws. The tests were completed during disc golf off-season, because of scheduling reasons. Furthermore, the tests were completed one by one, and the test times were chosen based on the suitability for the participants. There were no standardised guides what the participants could or could not do before the test, because the test was classified as non-encumbering, thus they did not need to have a certain resting time. However, they were all advised to bring their own putters and to wear appropriate sports clothing, which included shorts or pants that had pockets for the smartphone to be held during the putting.

The actual test consisted of 25 throws in total. Five throws were done per different distance, which were 5, 10, 15, 20 and 25 meters measured from the basket. All the throwing spots were in a perpendicular line from the basket. These distances were decided based on usual disc golf putting distance. However, the 25-meter distance was deemed to be too far and irrelevant after the tests and the data was not included in the final analysis. Thus, only the distances known in disc golf terms as circle one (0-10 meters) and circle two (10-20 meters) were included in the analysis. All the designated throwing spots were marked with a cone and the participant was advised to throw from the left side of the cone. As is in disc golf official rules, the participants had to maintain their balance and not move forward from the throwing spot before the disc had

landed, in the circle one distances. From further distances they were allowed to move forwards after the throw.

Participants started from the five-meter putting spot and moved further after each five throws. Before a throw they needed to put the video recording on from the official application (Invisible Companion App) designed for the glasses on a smart phone connected to the glasses. After the putt they needed to also stop the recording themselves and save it to the cloud of the equipment provider. This was designed on purpose so that the putting situation would be more like a competition situation and the participant could not complete multiple putts too rapidly in a row, much like in a competition. Furthermore, the recordings of each putt needed to be separated for analysing purposes. The pause time in-between throws would vary between a few minutes, depending on the functionality of the phone's application and the gaze glasses. If an error occurred during the recording and the putt was not recorded, because of the failure of the equipment, the throw would be taken again, despite the result of the throw.

Before the participant was allowed to putt, they were asked if they were ready. Once confirmed to be ready, the participant was given permission to putt. After every throw the participant was asked to give their subjective opinion whether the putt was successful or not in their own opinion. Only a yes or no answer was asked. This was for the purpose of later analysis. The result of the throw and opinion of success were recorded by the scribe (attachment 1). The test environment was made as quiet as possible, and no other people were allowed in the test environment. However, this was not possible during two professional player tests. The participants were advised to not rush in their putting and to perform as they would be in a competition. Furthermore, complete silence was required from all the testers, and no additional feedback or motivation was given to the participants during or after the putts. The participants were not told any in-depth details from what was tested, only that it is to do with gaze during putting. This was so that the participants would not focus too much on the gaze and rather would putt as they are used to in a non-test environment. The total test time varied from 30 to 45 minutes, again, depending on the functionality of the test equipment.

6.3 Data collection equipment

The participants were advised to use their own putters during the test for familiarity reasons. The disc golf baskets used were all ones used in professional environments and courses. The baskets used were either from brands Discmania or Prodigy. The equipment used to record can be seen from figure 8. The gaze of the participants was recorded with gaze glasses model Pupil Labs Invisible. These glasses were attached with a USB-C cable to a smartphone model OnePlus 8T, which is the official model approved for the glasses for optimal performance. (figure 8)

FIGURE 8. Equipment used to record the gaze of the participants.

The glasses have two cameras to record the performance of the participant as seen in figure 9. The camera that records the movement of the eyes is located on the inside wall of both sides of the glasses. The second camera, which records the visual field of the participant, is on the left side of the glasses. (figure 9) This scene camera also has an integrated microphone to record sounds. The scene camera records in a resolution of 1080 p and 30 frames per second (fps). The eye tracking camera records with a speed of 200 fps. (Pupil Labs 2025)

FIGURE 9. Pupil Labs Invisible glasses technical details. (Pupil Labs 2025)

6.4 Data collection and analysis

The data recorded with the glasses was immediately sent to the official Pupil Cloud, where the data could be later analysed. The video in the cloud shows where the person is looking at. As can be seen from figure 10, the red circle indicates the current gaze position, the number next to it indicates the number of current fixation and the blue circle indicates the gaze offset in the Companion App (figure 10).

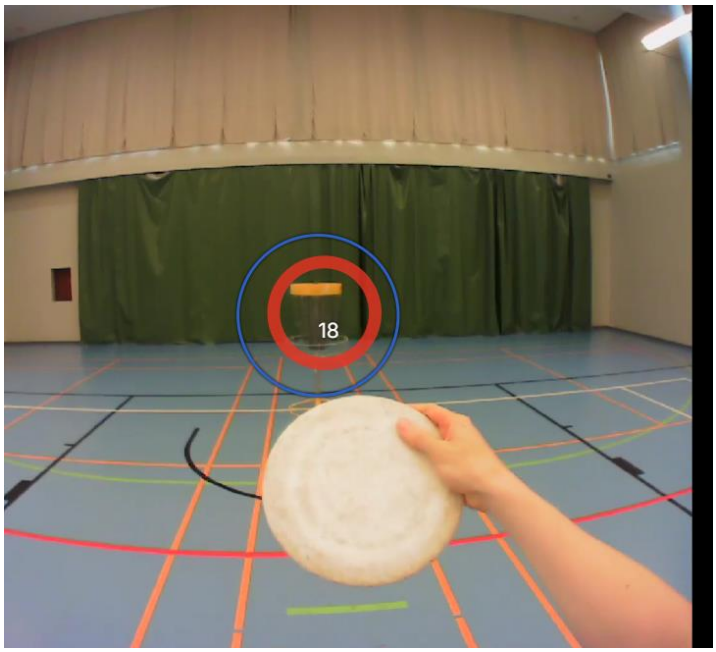

FIGURE 10. Gaze of the participant recorded from the 5-meter putting distance.

All the data was then labelled per participant in the official cloud service to ease the analysis of each participants' results. The data was exported from the cloud to an Excel format. The extraction provides multiple different Excels, however, only the one, which had the amount and the durations of each fixation, was used. Each video (n. 160) was watched simultaneously when its corresponding data was analysed from the Excel. The data analysed from the video and Excel were the following. The fixation in which the disc left the participant's hand and the duration (ms) of the said and previous fixation were the ones used. The position of the gaze whether it was on the target (basket) or off the target was analysed. Furthermore, the number of fixations between the putter leaving the participant's hand and landing either in the basket or on the ground was used for analysis and the time between the used fixations was measured for later analytical purposes. Finally, the success of quiet eye was measured. The success was defined as having the gaze position on target and lasted for over 100 ms when the putter left the participant's hand. However, if the duration was less than 100 ms or the gaze was off target, the quiet eye was unsuccessful on the said putt. These results were compiled on a results table on Excel (attachment 2), which also included the results of the putt and perceived success of the putt. Furthermore, three spit-outs were recorded during the tests. In disc golf terms spit-out means for the putt, which initially seems to go in the basket, results in coming out of the basket. This can be often because of high putting speed or a hit on the middle pole. These spit-outs were recorded as a miss, because of analysis purposes. Although different data was recorded and analysed after the tests, only data and results, which are relevant to the hypothesis were used on this thesis.

The data was analysed using the IBM SPSS Statistics software. Two different analysis methods were used for different variables. First, the durations of fixations and the relevance on either the result of the putt or the perceived result were analysed using the independent samples t-test. Second, the results and relevance on results of all other variables were analysed using the crosstabs chi square analysis. The significance of skill level was also analysed on each variable.

6.5 The reliability of the research and ethical issues

There is no previous gaze target or quiet eye research on the sport of disc golf, therefore all the data mentioned previously was analysed to see, which variables have relevance on results.

However, there are multiple meta-analysis about quiet eye, which variables have previously shown the most relevance on results and which ones explain the skill level differences between participants. These were used as guidelines when conducting the analysis. For example, Klostermann and Moeinirad (2020) proved in their meta-analysis that the most relevant variables to explain the skill difference were the duration of quiet eye, gaze location and the number of fixations.

There were eight participants in total, from which four were classified as professionals and four as amateurs. Therefore, it is worthy to mention that the sampling unit size is small, even though the number of putts analysed from each distance was 40 and in total 160.

The reliability of the gaze recording equipment has been tested previously by the manufacturer. According to Tonsen et al. (2020), the Pupil Invisible glasses have a reliable functionality independent of lighting conditions, rapid head movements, gender or age of the wearer as well as the use of contact lenses. The study, which included 367 participants of random age and gender, found that the glasses have more error on horizontal and lower part of the field of view than vertical and upper part of field of view. The latter could be because of eye lashes obstructing the view when looking down. Furthermore, the glasses were found to have some bias on estimating the gaze more centrally. The highest mean sample error was calculated in the lower quarter of the field of view an error of approximately 6.5 degrees. However, this is an overestimation of the total random spread across the field of view, which was evaluated by the researchers to be less than 4 degrees, thus giving a fairly accurate estimation of the gaze. (Tonsen et al. 2020)

The study has been designed, conducted and analysed in an ethical way following the official guidelines of The Finnish National Board on Research Integrity TENK. Furthermore, for the sake of reliability and credibility of this thesis, as many sources as possible were used in the literature review and have been referenced in an ethical manor.

7 RESULTS

The results segment will go through the results of the putts as well as the results of each research question. The overall putting results and the perceived (Exp result) results of the tests can be seen from figure 11 per skill level and distance.

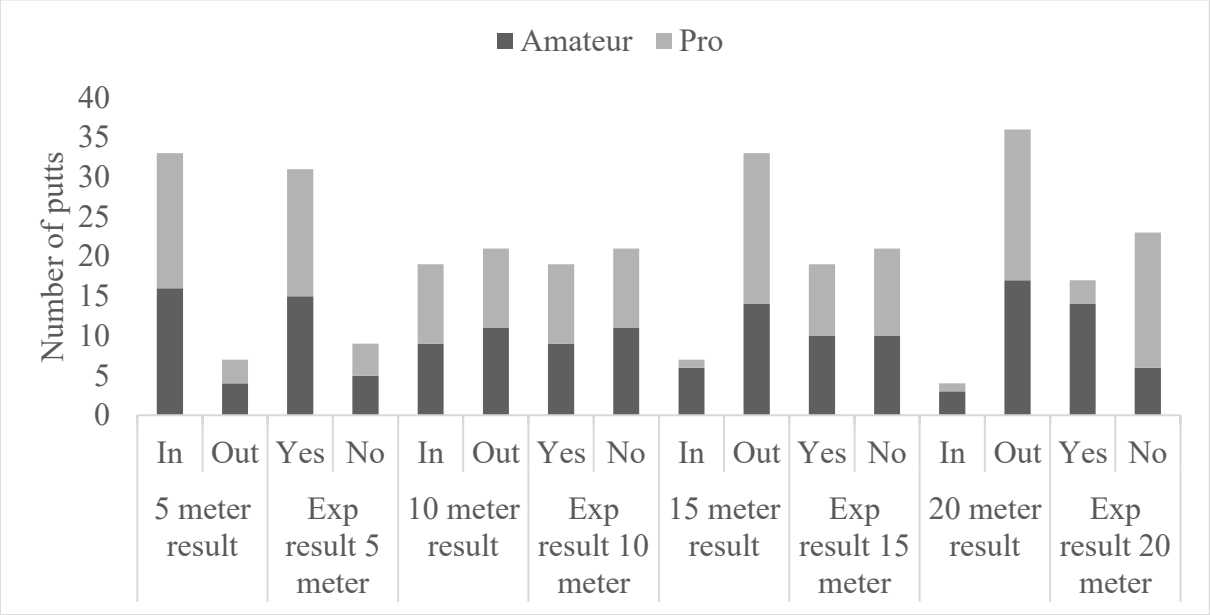

FIGURE 11. Putt results and perceived results of participants per level. The professionals' results are the upper part of the bars with lighter colour and the amateur results are the darker colour bars.

7.1 Research question 1 results

Gaze control was an important variable on this study. The results of gaze control during putting per level can be seen from figure 12.

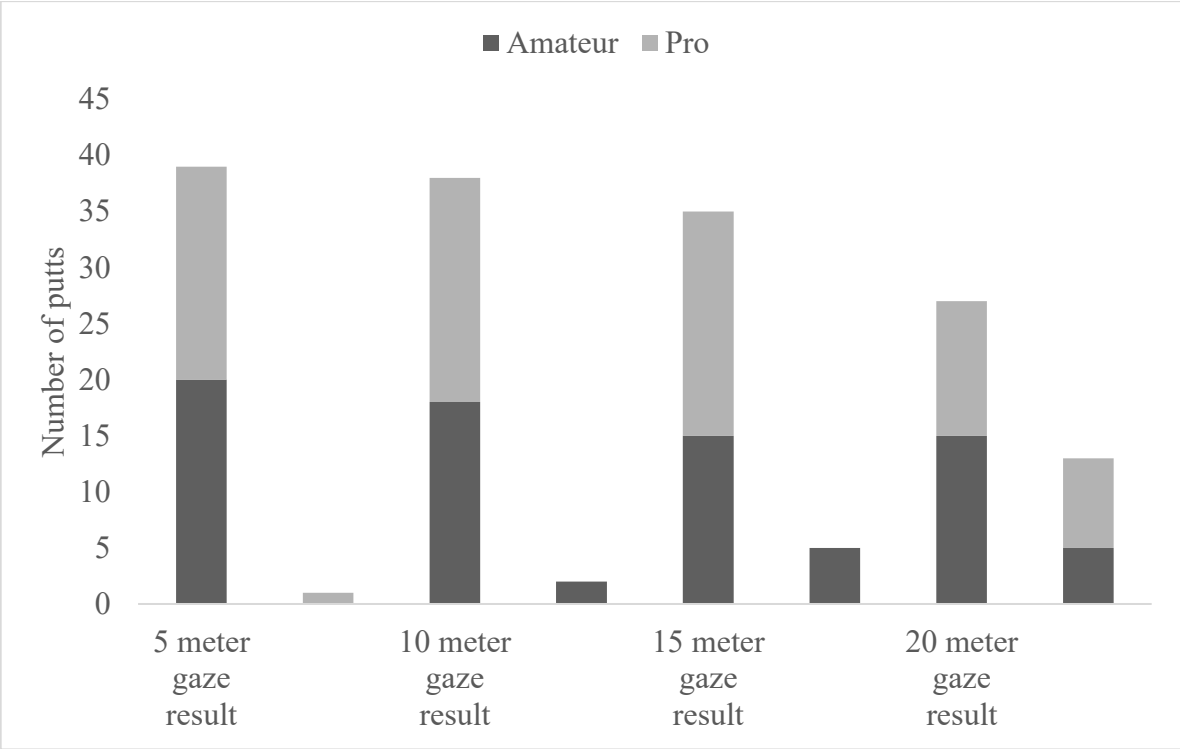

FIGURE 12. Gaze control results of participants per level. The left bar of each distance indicates the putts that have gaze on the target. Right bars of each distance indicate the off-target gazes. The lower part of the bars with the darker colour is the amateur results and the lighter colour the professionals.

The significance of gaze control on putting results and perceived results was again analysed using SPSS crosstabs Pearson Chi-Square analysis. The results are compiled in table 1. Again, the p-value needed to be below 0.05 to have significance. (table 1)

TABLE 1. Gaze target significance on putting result and perceived result.

Distance (meters)	Pearson Chi-Square value
5 m result	p = 0.028*
5 m perceived result	p = 0.060
10 m result	p = 0.168
10 m perceived result	p = 0.168
15 m result	p = 0.875
15 m perceived result	p = 0.188
20 m result	p = 0.854
20 m perceived result	p = 0.135

* $p < 0.05$

To understand better the gaze control of both skill levels, the number of fixations were also analysed between the putter leaving the hand of the participant and landing either to the basket or on land per distance. This averaged data can be seen from table 2. Both, putting result and perceived results, are included in the analysis. (table 2)

TABLE 2. Average fixation amount between the release and landing of the putter per level.

Level	Distance (meters)	In (Avrg. Fixation amount)	Out (Avrg. Fixation amount)	Successful exp result (Avrg. Fixation amount)	Unsuccessful exp result (Avrg. Fixation amount)
A	5 m	0,19	1,50	0,13	1,40
A	10 m	1,67	1,55	1,67	1,55
A	15 m	2,50	3,07	2,60	3,20
A	20 m	1,33	3,29	3,00	3,00
P	5 m	0,35	0,33	0,38	0,00
P	10 m	1,50	1,90	1,60	1,80
P	15 m	2,00	2,21	2,22	2,18
P	20 m	2,00	3,23	2,33	3,33

The total average amount of fixations between the putter leaving the hand of the participant and landing either to the basket or on land per distance was also calculated and this data is displayed on table 3. This data is not divided per skill level. (table 3)

TABLE 3. Total average fixation amount between the release and landing of the putter.

Distance (meters)	In (Avrg. Fixation amount)	Out (Avrg. Fixation amount)	Successful exp result (Avrg. Fixation amount)	Unsuccessful exp result (Avrg. Fixation amount)
5 m	0,24	1	0,26	0,78
10 m	1,58	1,71	1,63	1,67
15 m	2,43	2,58	2,42	2,67
20 m	1,5	3,26	2,88	3,24

The significance of amount of fixations between the release and landing of the putter on putting results and perceived results was calculated using the SPSS crosstabs Pearson Chi-Square analysis. The significance results are displayed on table 4. The needed p-value for a variable to be significant was below 0.05. (table 4)

TABLE 4. Amount of fixations between the release and landing of the putter significance on putting results and perceived results.

Distance (meters)	Pearson Chi-Square P-value
5 m result	$p < 0.001$*
5 m perceived result	$p = 0.003$*
10 m result	$p = 0.156$
10 m perceived result	$p = 0.917$
15 m result	$p = 0.151$
15 m perceived result	$p = 0.323$
20 m result	$p = 0.316$
20 m perceived result	$p = 0.262$

* $p < 0.05$

The significance between amount of fixations between the release and landing of the putter and skill level was calculated using the SPSS crosstabs Pearson Chi-Square analysis. The significance results can be seen on table 5. Likewise, previously, the needed p-value for a variable to be significant was below 0.05. (table 5)

TABLE 5. Amount of fixations between the release and landing of the putter significance on skill level.

Distance (meters)	Pearson Chi-Square P-value
5 m level	p = 0.135
10 m level	p = 0.074
15 m level	p = 0.038*
20 m level	p = 0.729

* $p < 0.05$

7.2 Research question 2 results

One of the key data in this test was the fixation durations during putting, which can be seen from table 6. The results are in milliseconds (ms) and are displayed per test participant – amateurs (A) and professionals (P). The results have been averaged from the total four putts per distance per participant.

TABLE 6. Average fixation duration during putting per participant per distance.

Participant	Average Fixation duration (ms) during putting, 5 meters	Average Fixation duration (ms) during putting, 10 meters	Average Fixation duration (ms) during putting, 15 meters	Average Fixation duration (ms) during putting, 20 meters
A1	4408	5287	997	524
A2	11787	13133	11059	6507
A3	4519	7823	8735	12304
A4	9139	4608	1720	104
P1	4572	99	2539	482
P2	2392	2485	1240	2065
P3	6421	5028	3814	3881
P4	6739	5839	2810	1372

The results of average fixation duration during putting can be inspected in a more graphical way with the standard deviation bars from figure 13. The results are shown per test participant and colour coded per distance. (figure 13)

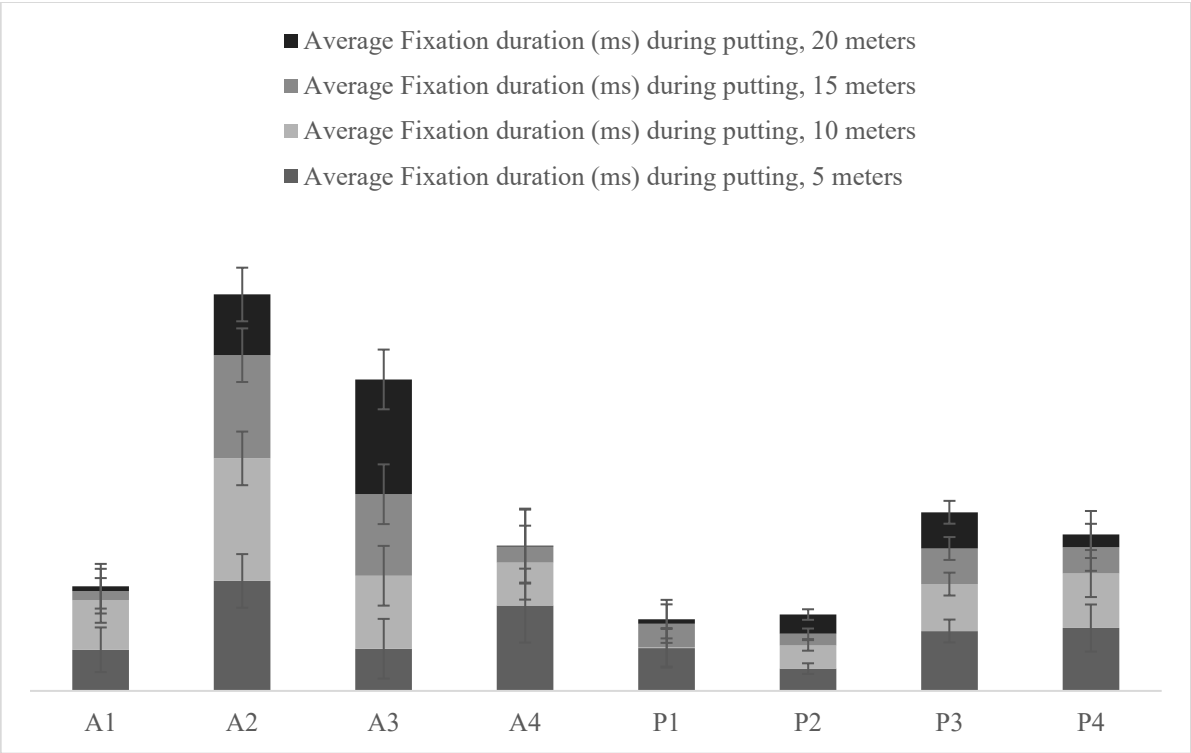

FIGURE 13. Average fixation duration per test participant with the standard deviation bars.

The total average fixation duration during putting from all the putts combined per level was also calculated. The amateurs had an average of 6416 ± 5838 ms and the professionals 3236 ± 2272 ms.

7.3 Research question 3 results

The significance of fixation duration, during putting, on putting results and perceived results were another key information to analyse in this research. These results are shown in table 7. For a fixation duration to have significance on putting or perceived result, they needed to have a p-value lower than 0.05 (table 7). This data was analysed using the SPSS independent samples t-test.

TABLE 7. The significance of fixation duration, during putting, on putting results and perceived results.

Distance (meters)	P-value
5 m result	0.663
5 m perceived result	0.826
10 m result	0.559
10 m perceived result	0.850
15 m result	0.621
15 m perceived result	0.595
20 m result	0.823
20 m perceived result	0.153

p* < 0.05

The final variable analysed in this study was the quiet eye (QE). These results were also divided between the amateurs and professionals and per distance, which can be seen from figure 14. The amateurs are the lower part of the bar and the results of the professionals the upper part of the bars with the lighter colouring. (figure 14)

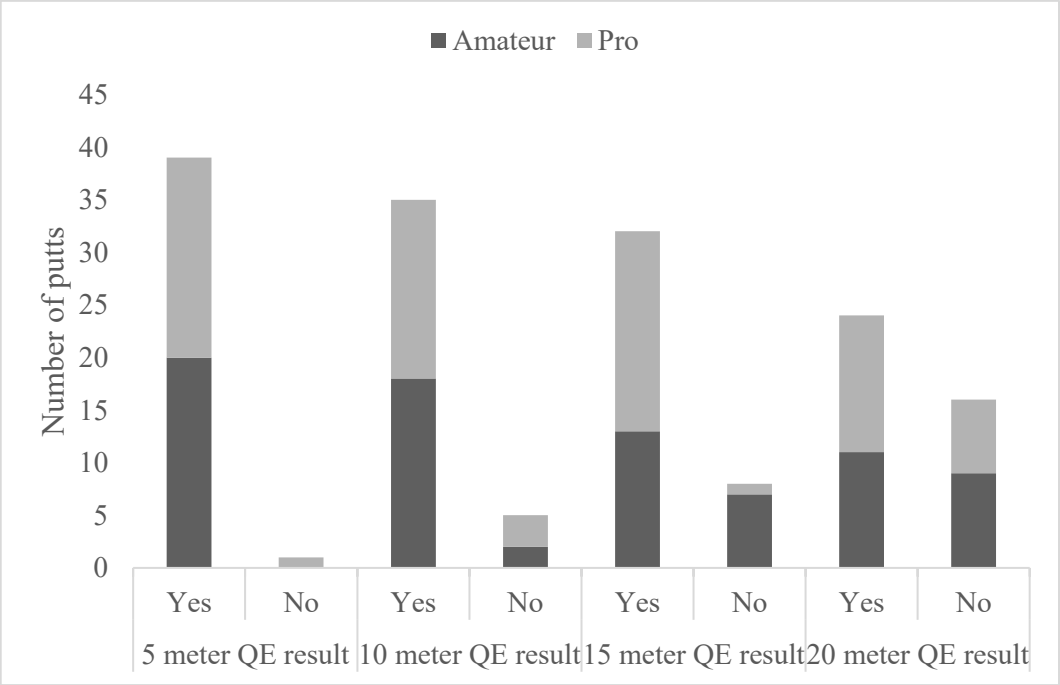

FIGURE 14. Quiet eye results per level per distance.

Lastly, the significance of quiet eye success on putting and perceived results were analysed using the same analysis methods as previously with the gaze control relevance. The results are displayed in table 8. The needed p-value for a result to have significance had to be below 0.05. (table 8)

TABLE 8. Quiet eye relevance on putting and perceived results.

Distance (meters)	Pearson Chi-Square P-value
5 m result	p = 0.028*
5 m perceived result	p = 0.060
10 m result	p = 0.550
10 m perceived result	p = 0.550
15 m result	p = 0.533
15 m perceived result	p = 0.527
20 m result	p = 0.604
20 m perceived result	p = 0.064

* $p < 0.05$

8 DISCUSSION

The purpose of this study was to find, if there are differences in gaze control and quiet eye (QE) between amateur and professional disc golfers in a putting environment. Furthermore, this study was conducted to bring more knowledge in the field of disc golf science. There were three research questions this study wanted to try to find answers to. The first question was, if the professionals have a better gaze control than amateurs. The second was, if the duration of QE changes when the putting distance changes. The final hypothesis was, if the duration and success of QE have a correlation on the success of putting. The study setting was amateurs versus professionals, and the data is mostly analysed by level. The main findings of this study were that the two skill levels did not have major differences between each other in gaze control nor in quiet eye. However, the success of quiet eye and the amount of fixations between the putter leaving the participant's hand and landing, did have significance on the results.

8.1 The conclusions of the research questions

Before dissecting other results connected to the hypothesis, it is important to go through the actual putting and perceived results. The results were somewhat surprising, since the amateurs had better results, especially from longer distances than the professionals (figure 11). The amateurs had a total of 34 putts in compared to the professionals 29. Furthermore, the amateurs also had more successful perceived results, 48 compared to the professionals' 37. This could indicate that the amateurs felt more successful compared to their own skill level than the professionals, since they had only eight more perceived successful putts whereas the amateurs had 14. Furthermore, in a few cases it was noted that if there were unsuccessful putts early in the test, it might have had some impact on the self-confidence, which then might have reflected to the putting and perceived results, which could have led to choking under pressure (Baumeister 1984, 610). If a motivational intervention had been conducted during the tests, it could have helped the participants to perform better. A motivational intervention can increase the perception of emotional and cognitive control, which can lead to increased confidence. (Hennessey 2022)

8.1.1 Research question 1 conclusions

According to the results, objectively, it can be said that the first hypothesis is untrue, since there were no significant differences between the two skill levels in gaze control. The second hypothesis can be said to be true, since the QE duration changes with the distance no matter the skill level. Finally, the third hypothesis can be said to be mostly untrue, because there was zero significance on the duration of fixations on the putting results and the success of QE only had significance on one distance group.

When looking at the results of the gaze targeting (figure 12) the results were very similar for professionals and amateurs. However, only in the distance of five meters did the amateurs have all putts gazed on-target. The professionals, however, had all putts gazed on-target on 10- and 15-meter distances and all but one on-target on five meters. These results did not, however reflect on the results of actual putting nor on the perceived results. Only, from the five-meter distance, there was a significance on the gaze control with the results ($p = 0.028$). This means that, it was significant on the putting results, if the participant had their gaze on or off the target. There was no other significance on gaze targeting on the results. Furthermore, when looked at the actual five-meter putting results, the professionals had 17 out of 20 successful putts, whereas the amateurs had 16. The difference is only one and the professionals had one less gaze on-target than amateurs. One reason for these non-significant results could be the low number (n. 40) of putts per distance in total.

The gaze control was also analysed using the amount of fixations in-between the putt leaving the hand and landing either in the basket or on the ground. The results show that putts that were successful and perceived as successful had less fixations than the ones that were unsuccessful. This was mostly the case with both skill levels and when the average was calculated totally the successful cases had always less fixations than the unsuccessful ones (table 3). For example, from the 15-meter distance, amateurs had an average of 2.5 fixations on the successful putts, whereas the unsuccessful ones had more than 3. Similar results were recorded with professionals. The professionals had an average of 2.0 fixations on the successful putts and the unsuccessful ones had more than 3 (table 2). Although only the five-meter distance results had a clear significance between the number of fixations and success of results ($p > 0.001$) and perceived success ($p = 0.003$), it can be said that less fixations were probably linked to better success. This could be, because the participant is able to focus better on the target and being

less prone to external disrupts and this can lead to a better result even when under pressure (Lebeau et al. 2016).

Similar results have been reached in previous gaze-studies, for example, when Reunanen (2024) researched the differences in gaze control between professional and amateur javelin throwers. The results showed that professionals and amateurs had no significant difference in the number of fixations between the cross-step and throw.

Although there was significance only on one distance between the gaze control and the putting results, this data was nevertheless informative. When the significance between skill level and fixation amount was analysed (table 5), there was significance on the 15-meter distance (0.038). This significance means that a higher skill level is related to lesser amount of fixations. This can be the result of target throwing practice, because once focus is maintained there will be less eye movement, therefore less fixations and again could lead to better results (Vickers 2007, 68).

When looked in a more subjective view, in most cases where the gaze was off target, for example on the right side of the target, the putt was more likely to miss from the right side. Furthermore, the likeliness of a putt missing the target when the gaze was not on target was higher, almost 90 % (attachment 2). To conclude the first hypothesis, although there were no differences between professional and amateur gaze control, there was a connection with fewer fixations and on-target gaze to lead more likely to successful putts.

8.1.2 Research question 2 conclusions

The QE duration was analysed from the fixation where the putter left the participant's hand. This was similar to the research of Giancamilli et al. (2022) where the onset of QE in basketball free-throw was defined to be when the throwing arm is extended to the throw (Giancamilli et al 2022). The QE durations recorded had major changes with the distance changes. When looked at the results one can see that in most cases the QE duration decreases with the distance. Furthermore, only one participant had a higher QE duration from the 20-meter distance than from the 5-meter (table 6). This could be mainly due to the higher difficulty of putting from longer distances. Although many studies support the fact, that increased difficulty in tasks could lead to longer QE durations, especially among professional athletes (Ziv & Lidor 2019). Also,

the longer distances are not as familiar to putt from as are the shorter distances, therefore it can be harder to maintain focus, which affects the duration of the QE. Furthermore, less training from a longer distance can increase the difficulty of knowing where to target and how to control the gaze (Vickers 2007, 10).

The two skill levels also had differences between them. On average the amateurs had a longer QE duration (6416 versus 3236 ms) than the professionals, but the standard deviation was also much higher (5838 versus 2272 ms) with the amateurs. The QE duration with the professionals had therefore less spread than the amateurs. The results are contrary to the meta-analysis results of Klostermann and Moeinirad (2020) where 12 studies had found that professionals have longer QE duration than amateurs and 14 where the result is non-significant. Only a few studies reported a negative result. (Klostermann & Moeinirad 2020) However, the results follow the meta-analysis of Gegenfurtner et al. (2011), where they enclosed that a shorter fixation duration could reflect on experts' capability of faster encoding and information retrieving, in other words faster information processing (Gegenfurtner et al. 2011). Furthermore, Klostermann et al. (2018) found that a QE duration of over 3000 ms does not bring increased benefits on the success of the targeting task (Klostermann et al 2018).

8.1.3 Research question 3 conclusions

The relevance of both duration and success of QE on the success of putting and perceived results was the final research question of this research. The significance of duration of QE on putting results and perceived results were analysed from all distances and all the results were non-significant (table 7). Therefore, according to this research there is no significant correlation between the duration of QE and success of putting in either skill levels. Which, for example, goes against the study of Vickers (2016), where it was proved that higher QE duration could lead to increased sports performances in targeting tasks.

A successful QE in this study was defined to have a duration of more than 100 ms and for the gaze to be on target and stay there until the putter left the participant's hand. The QE has been proved to showcase a higher level of sports expertise in multiple research (Vickers, 2007, 11). Therefore, it was important to use this variable also on this research. As the results shows, only in the 5-meter distance did the success of QE have significant relevance on the putting success

(table 8). Furthermore, the longer the distance the more there were unsuccessful QE phenomena. This was especially true with amateurs, whereas the professionals could maintain the QE more successfully even with further distances (figure 12). This can be again because of previous training of the gaze control and being more familiar with longer distance putting (Vickers 2007, 11).

8.2 Study limitations

There are multiple limitations that need to be accounted for in this study. Firstly, as stated earlier in the discussion, the sample size is quite small. This might affect the significance results, and it does not have enough variation between participants. Furthermore, the setting was professional versus amateur, however the level of skill the amateurs had was probably closer to intermediate than amateur, because of their multiple year experience.

Secondly, this was the first research the author has conducted solo. Therefore, it can be assumed that there were some errors among the analysis and original test setting. This was also the first putting research done on disc golf in 30 years and no reference material could be used to support this research. Hence, this research can be also seen as a pilot study. For example, the variable used "perceived result" was used with the idea of understanding better, if the participants are more satisfied with their putts, even though it would not go in. Putting is an extremely technical movement, and players have different techniques compared to others and even compared to the distance putted from. Therefore, the author wanted to include something to give understanding, if the participant was able to putt with their desired technique.

Thirdly, the equipment used on this study was somewhat outdated and had some issues. The Pupil Invisible Glasses only worked with a specific phone, which was used in the tests. However, what was not known was that the glasses were designed to work with a particular operating system, which was an outdated one. Hence, once the phone updated its operating system the glasses and its software did not function as well as could have and issued some minor and even major defects. For example, in a few cases the software had an error on the recording a few seconds after the recording had started, which caused the file to corrupt, and the data could not be used. Also, when A2 threw their last putt, it originally went in, but the software did not start the recording and needed to be rerecorded, in which the putt did not go

in. These issues had some impact on the results and on the test experience of the participants. One must add, that although the glasses have been designed to look and feel like typical eyeglasses (Tonsen et al. 2020), this could still have some unnatural feeling on the test participants, which might affect the putting.

Fourthly, the putting environment was designed to be free from external distractions as other people were not allowed in the putting environment. However, during the tests of P4, this was not possible as there were other professionals practicing putting. The participant said that this did not disturb him and that he was used to it. Other people were advised to be as quiet as possible during the test and to stay behind the participant so that his gaze would not be compromised during the tests. Although the participant stated that this did not distract him, it can be assumed that there was some effect. Furthermore, with participant A4, before their 9th throw there was a short disturbance caused by a door opening before the throw, which meant they needed to stop the throw and do it again after the space was free of disturbance again. The putt was the only one to not go in from the 10-meter distance, so there could very well be some impact on the focus for the participant. However, it needs to be stated that in a competitive putting environment, there can be multiple distractions from other players, crowd or natural environmental distractions such as weather or animals.

Finally, for many of the professional-levelled participants this was also a new situation to be in. All the amateur participants had some experience from research settings or from being a test subject, whereas most of the professionals did not for the knowledge of the study conductor. Nerves and an unfamiliar situation could have impact on both skill level participants. However, especially professionals are used to a competitive environment, where stress and nerves come into play often in putting, which could be assumed to translate into similar situations such as this study setting.

8.3 Future suggestions

Disc golf is a rapidly growing sport that needs more research. Therefore, this study can be used as a stepping stone for further putting research. There are multiple limitations that have been covered, but nevertheless this study serves its purpose in providing more knowledge of gaze control and quiet eye on a disc golf putting setting – a setting that has never been earlier

researched officially. This data and analysis can be used for coaching and putting skill developing purposes.

The author would recommend more in-depth future research on this matter with better technology, resources and sample size and variety. This variety can be for example an interventive study, where the subject is given either motivation or instructions. Putting is key to success in disc golf, therefore it is important to know the importance of gaze control in targeting tasks. Furthermore, from a biomechanical perspective, it would be imperial for the science of the sport to have movement analyses on putting researched. This would bring important data and knowledge for disc golfers and coaches.

Disc golfers of all skill level could benefit from knowing their strengths and weaknesses in gaze control in putting, therefore it could be extremely beneficial to do more research on this matter and including it more into coaching and training settings. Creating and using different gaze training methods can be assumed to be beneficial for increased putting skill, since these motor skills can be improved with proper training.

REFERENCES

Anttila, T., Karppi, K. (2016). Liikuntapaikkasuunnitelma 2029. Website. Cited on 15.11.2024. https://issuu.com/turunliikunta/docs/liikuntapaikkasuunnitelma_2029

Chen, C.-C. (JJ), Ryuh, Y., Luczak, T., & Lamberth, J. (2022). Examining Different Foci of Attention on Golf Putting Performance in Novice Learners. Physical Educator, 79(4), 466–483. doi.org/10.18666/TPE-2022-V79-I4-10828

Baumeister, R. (1984). Chocking Under Pressure: Self-Consciousness and Paradoxical Effects of Incentives on Skillful Performance. doi.org/10.1037/0022-3514.46.3.610

Bocks, D. (2019). Incidence of shoulder pain in professional and semi-professional disc golfers. Satakunta University of Applied Sciences. Thesis. Cited on 27.10.2024. https://www.theseus.fi/bitstream/handle/10024/166731/DanielaBocks.pdf?sequence=2&isAllowed=y

Dowling, B., McNally, M. P., Laughlin, W. A., & Onate, J. A. (2018). Changes in Throwing Arm Mechanics at Increased Throwing Distances During Structured Long-Toss. American Journal of Sports Medicine, 46(12), 3002–3006. doi: 10.1177/0363546518795892.

Enoka, R. M. (2015). Neuromechanics of human movement. Fifth edition. Champaign, IL: Human Kinetics.

Gegenfurtner, A., Lehtinen, E., & Säljö, R. (2011). Expertise Differences in the Comprehension of Visualizations: A Meta-Analysis of Eye-Tracking Research in Professional Domains. Educational Psychology Review, 23(4), 523–552. doi.org/10.1007/s10648-011-9174-7

Giancamilli, F., Galli, F., Chirico, A., Fegatelli, D., Mallia, L., Palombi, T., Cordone, S., Alivernini, F., Mandolesi, L., & Lucidi, F. (2022). When the going gets tough, what happens to quiet eye? The role of time pressure and performance pressure during basketball free throws. Psychology of Sport and Exercise, 58, 102057. doi.org/10.1016/j.psychsport.2021.102057

Giljarhus, K. E. T., Gooding, M. T., & Njærheim, J. (2022). Disc golf trajectory modelling combining computational fluid dynamics and rigid body dynamics. Sports Engineering, 25(1), 26. doi.org/10.1007/s12283-022-00390-5

Hennessy, S. L. (2022). Positive motivational self-talk: The influence of gender on performance, perception of self-talk, and performance anxiety during a disc golf putting exercise before and after a self-talk intervention. Eastern Washington University.

Master of Science in Physical Education. Master's thesis. Cited on 19.05.2025. https://dc.ewu.edu/theses/749/

Hummel, S. A. (2003). Frisbee Flight Simulation and Throw Biomechanics. University of California. Master of Science. Thesis. Cited on 25.10.2024. https://morleyfielddgc.wordpress.com/wp-content/uploads/2009/04/hummelthesis.pdf

Holcomb, L. (2023). EMG Activation of the Upper Extremity During a Standing Backhand Disc Golf Throw. East Tennessee State University. Sport, Exercise, Recreation and Kinesiology. Thesis. Cited on 12.12.2024. https://dc.etsu.edu/honors/820/

Jackson, S. L., & Tanner, E. A. (1993). Analysis of elite and novice disc golfers performing 10 m putts. https://research.ebsco.com/linkprocessor/plink?id=3e5a5032-2eca-38a8-8682-8e86341de220

Kamaruddin, N., Potts, J., & Crowther, W. (2018). Aerodynamic performance of flying discs. Aircraft Engineering and Aerospace Technology, 90(2), 390–397. doi.org/10.1108/AEAT-09-2016-0143

Karandikar, N., & Vargas, O. O. O. (2011). Kinetic Chains: A Review of the Concept and Its Clinical Applications. PM&R, 3(8), 739–745. doi.org/10.1016/j.pmrj.2011.02.021

Klostermann, A., Kredel, R., Hossner, E.-J., & Klostermann, A. (2018). Quiet Eye and Motor Performance: The Longer the Better? Journal of Sport & Exercise Psychology, 40(2), 82–91. doi.org/10.1123/jsep.2017-0277

Klostermann, A., & Moeinirad, S. (2020). Fewer fixations of longer duration? Expert gaze behavior revisited. German Journal of Exercise and Sport Research, 50(1), 146–161. doi.org/10.1007/s12662-019-00616-y

Koch, N., McGuire, H., Cole, C., & Lindsey, Z. (2024). The effects of thumb position on backhand disc golf throws of a mid-range disc. doi.org/10.1063/5.0229599

Laine, H. (2017). Venyttelyopas frisbeegolfin harrastajille liikkuvuuden kehittämiseen. Kajaani University of Applied Sciences. Thesis. Cited on 22.3.2025. https://urn.fi/URN:NBN:fi:amk-201705097013

Lebeau, J.-C., Liu, S., Sáenz-Moncaleano, C., Sanduvete-Chaves, S., Chacón-Moscoso, S., Becker, B. J., & Tenenbaum, G. (2016). Quiet Eye and Performance in Sport: A Meta-Analysis. doi.org/10.1123/jsep.2015-0123

Leppilahti, O.-P. (2019). Frisbeegolfin rystyheiton kinemaattinen analyysi. University of Jyväskylä. Faculty of Sports Sciences. Thesis. Cited on 08.11.2024. https://urn.fi/URN:NBN:fi:jyu-202002282219

Mann, D. T. Y., Williams, A. M., Ward, P., & Janelle, C. M. (2007). Perceptual-Cognitive Expertise in Sport: A Meta-Analysis. doi.org/10.1123/jsep.29.4.457

Nelson, J. T., Jones, R. E., Runstrom, M., & Hardy, J. (2015). Disc Golf, a Growing Sport: Description and Epidemiology of Injuries. Orthopaedic Journal of Sports Medicine. doi.org/10.1177/2325967115589076

O'Neill, R. (2022, November 23). The Disc Golf Boom – Let's Look at the Numbers. Innova Disc Golf. Website. Cited on 19.11.2024. https://www.innovadiscs.com/innova-news/the-disc-golf-boom-lets-look-at-the-numbers/

PDGA (2023). Yearend demographics report. Website. Cited on 18.11.2024. https://www.pdga.com/files/2023_pdga_yearend_demographicsreport.pdf

Pupil Labs (2025). Technical overview. Website. Cited on 13.02.2025. https://docs.pupil-labs.com/invisible/

Putnam, C. A. (1993). Sequential motions of body segments in striking and throwing skills: Descriptions and explanations. Journal of Biomechanics, 26 Suppl 1, 125–135. doi.org/10.1016/0021-9290(93)90084-r

Reunanen, S. (2024). Keihäänheittäjän katseenkäyttäytyminen heittosuorituksen aikana. University of Jyväskylä. Faculty of Sports Sciences. Master's thesis. Cited on 05.01.2025. https://urn.fi/URN:NBN:fi:jyu-202406054277

Risto, T. (2023). Ihmisen fysiologiaa: Perusteista pidemmälle, Lahti: VK-Kustannus.

Schmidt, R. A., Lee, T. D., Winstein, C., Wulf, G., & Zelaznik, H. N. (2018). Motor Control and Learning: A Behavioral Emphasis. Sixth edition. Champaign, IL: Human Kinetics.

Serrien, B., & Baeyens, J.-P. (2018). Systematic Review and Meta-Analysis on Proximal-to-Distal Sequencing in Team Handball: Prospects for Talent Detection? Journal of Human Kinetics, 63(1), 9–21. doi.org/10.2478/hukin-2018-0002

Finnish Disc Golf Association (2024). Disc Golf in Finland—Statistics. Website. Cited on 15.11.2024. https://frisbeegolfliitto.fi/disc-golf-in-finland-statistics/

Tervonen, A. (2022). Olkanivelen ulko- ja sisäkierron sekä selkärangan kierron yhteys frisbeegolfin rystyheiton heittonopeuteen. Tampere University of Applied Sciences. Thesis. Cited on 22.03.2025. http://www.theseus.fi/handle/10024/783886

Tonsen, M., Baumann, C. K., & Dierkes, K. (2020). A High-Level Description and Performance Evaluation of Pupil Invisible. doi.org/10.48550/arXiv.2009.00508

UDisc. (2023) Disc Golf Growth Report powered by UDisc. Website. Cited on 15.11.2024. https://udisc.com/disc-golf-growth-report

Vickers, J. N. (1996). Visual control when aiming at a far target. Journal of Experimental Psychology: Human Perception and Performance, 22(2), 342–354. doi.org/10.1037/0096-1523.22.2.342

Vickers, J., & Williams, A. (2007). Performing Under Pressure: The Effects of Physiological Arousal, Cognitive Anxiety, and Gaze Control in Biathlon. Journal of Motor Behavior, 39, 381–394. doi.org/10.3200/JMBR.39.5.381-394

Vickers, J. N. (2007). Perception, cognition, and decision training: The quiet eye in action. Human Kinetics. Champaign, IL: Human Kinetics

Vickers, J. N. (2016). Origins and current issues in Quiet Eye research. Current Issues in Sport Science (CISS), 1, 101–101. doi.org/10.15203/CISS_2016.101

Wagner, H., Pfusterschmied, J., Tilp, M., Landlinger, J., von Duvillard, S. P., & Müller, E. (2014). Upper-body kinematics in team-handball throw, tennis serve, and volleyball spike. Scandinavian Journal of Medicine & Science in Sports, 24(2), 345–354. doi:10.1111/j.1600-0838.2012.01503.x.

Walters-Symons, R., Wilson, M., Klostermann, A., & Vine, S. (2018). Examining the response programming function of the Quiet Eye: Do tougher shots need a quieter eye? Cognitive Processing, 19(1), 47–52. doi.org/10.1007/s10339-017-0841-6

Xu, S., Sun, G., & Wilson, M. R. (2021). Neurophysiological evidence of how quiet eye supports motor performance. Cognitive Processing, 22(4), 641–648. https://doi.org/10.1007/s10339-021-01036-3

Ziv, G., & Lidor, R. (2019). Gaze Behavior in Golf Putting—A Review. International Journal of Golf Science, 9(1), 1–52. https://www.golfsciencejournal.org/article/10140-gaze-behavior-in-golf-putting-a-review

Attachments

Attachment 1. Example of results of the test recorded

Pro 1 Gender: Male Age: 26 Height: 183cm Weight: 78,5kg

Distance (5 m)	Result (In (1)/Spit-out (2)/Out (0))	Perceived success on putt (Yes (1) /No (0))
	1	1
	1	0
	1	1
	1	1
	1	1

Distance (10 m)	Result (In/Spit-out/Out)	Perceived success on putt (Yes/No)
	1	1
	1	0
	1	0
	1	1
	1	1

Distance (15 m)	Result (In/Spit-out/Out)	Perceived success on putt (Yes/No)
	0	0
	0	1
	0	0
	0	0
	0	0

Distance (20 m)	Result (In/Spit-out/Out)	Perceived success on putt (Yes/No)
	0	0
	0	0
	0	0
	0	1
	1	0

Attachment 2. All the results of the test recorded on a single Excel file.

	Taso	Matka 5m tulos (In 1, Out 0, Spit out 2)	Koettu tulos	Gaze target (Basket 1, off target 0)	Amount of fixations between release and landing	QE duration before release (ms)	Time between previous and release fixation, more than 1s (Yes 1, No 0)	QE duration (ms) during release	QE (yes 1, no 0)	Matka 10m tulos	Koettu tulos	Gaze target (Basket 1, off target 0)	Amount of fixations between release and landing	QE duration before release (ms)	Time between previous and release fixation, more than 1s (Yes 1, No 0)	QE duration (ms) during release	QE (yes 1, no 0)
Ama1	1	0	0	1	2	553	1	4201	1	0	0	1	0	6777	0	1105	1
	1	0	0	1	2	4973	0	657	1	0	0	1	1	5057	0	329	1
	1	1	1	1	0	5121	0	1053	1	0	0	1	2	6009	0	157	1
	1	1	1	1	1	1746	1	6897	1	1	1	1	1	1517	1	12385	1
	1	1	1	1	0	345	1	9233	1	0	0	1	1	149	1	12457	1
Ama2	1	1	1	1	0	89	1	7861	1	0	0	1	1	3657	1	10797	1
	1	1	1	1	0	313	1	11257	1	1	1	1	0	177	1	19545	1
	1	1	1	1	0	345	1	12837	1	0	0	1	1	11601	1	5609	1
	1	0	1	1	0	161	1	14181	1	1	1	1	0	169	1	13233	1
	1	1	1	1	0	261	1	12801	1	0	0	1	0	233	1	16481	1
Ama3	1	0	0	1	2	7856	0	249	1	0	0	0	3	12477	0	73	0
	1	1	1	1	0	7249	0	1069	1	1	1	1	2	1485	1	11905	1
	1	1	0	1	0	12513	0	4025	1	0	0	0	3	10352	0	137	0
	1	1	1	1	0	12289	0	1485	1	1	1	1	3	629	1	13281	1
	1	1	0	1	1	149	1	15769	1	0	0	1	3	73	1	13721	1
Ama4	1	1	1	1	0	253	1	4609	1	1	1	1	2	7821	0	157	1
	1	1	1	1	0	89	1	11241	1	1	1	1	3	N/A	1	8256	1
	1	1	1	1	0	297	1	11673	1	1	1	1	2	121	1	4661	1
	1	1	1	1	1	101	1	8613	1	0	0	1	2	209	1	9829	1
	1	1	1	1	0	461	1	9557	1	1	1	1	2	6757	0	137	1
Pro1	2	1	1	1	0	145	1	2269	1	1	1	1	2	6217	0	93	0
	2	1	0	1	0	117	1	5209	1	1	0	1	2	4441	0	105	1
	2	1	1	1	0	221	1	4601	1	1	0	1	2	85	0	101	1
	2	1	1	1	0	165	1	5301	1	1	1	1	1	4473	0	105	1
	2	1	1	1	0	321	1	5481	1	1	1	1	2	4005	0	93	0
Pro2	2	1	1	1	1	1781	1	1705	1	1	1	1	1	1273	1	2938	1
	2	1	1	1	1	914	1	2824	1	1	1	1	2	698	1	2756	1
	2	1	1	1	1	598	1	2419	1	1	1	1	1	818	1	3273	1
	2	1	1	1	1	890	1	2467	1	0	0	1	0	2054	0	1698	1
	2	0	1	1	1	729	1	2547	1	1	1	1	0	2882	0	1760	1
Pro3	2	1	1	1	1	5027	0	2790	1	0	1	1	4	722	1	6072	1
	2	1	1	1	0	5667	1	6271	1	0	0	1	4	302	1	5353	1
	2	1	1	1	0	525	1	6731	1	0	0	1	4	353	1	6125	1
	2	1	1	1	0	209	1	9431	1	1	1	1	2	6261	0	65	0
	2	1	0	1	0	1035	1	6880	1	0	0	1	2	169	1	7525	1
Pro4	2	0	0	0	0	553	1	10205	0	0	1	1	1	7849	1	1157	1
	2	0	0	1	0	509	1	6289	1	0	0	1	0	561	1	9441	1
	2	1	1	1	0	1037	1	6437	1	0	0	1	1	593	1	8113	1
	2	1	1	1	0	2861	1	3645	1	0	0	1	1	2741	1	5113	1
	2	1	1	1	0	1121	1	7117	1	0	0	1	2	2257	1	5369	1

Matka 15m tulos	Koettu tulos	Gaze target (Basket 1, off target 0)	Amount of fixations between release and landing	QE duration before release (ms)	Time between previous and release fixation, more than 1s (Yes 1, No 0)	QE duration (ms) during release	QE (yes 1, no 0)	Matka 20m tulos	Koettu tulos	Gaze target (Basket 1, off target 0)	Amount of fixations between release and landing	QE duration before release (ms)	Time between previous and release fixation, more than 1s (Yes 1, No 0)	QE duration (ms) during release	QE (yes 1, no 0)
0	0	1	5	7845	0	181	1	0	0	1	3	5189	0	169	1
0	0	1	4	7001	0	329	1	0	1	1	6	3677	0	125	1
1	1	1	5	4129	0	85	0	0	0	1	5	8153	0	209	1
1	1	1	5	11477	0	229	1	0	1	1	3	4873	0	369	1
0	0	1	3	4249	1	4161	1	0	1	1	7	197	1	1749	1
0	1	1	0	5313	1	13465	1	1	1	1	0	13681	0	3701	1
1	1	1	0	241	1	15821	1	0	1	1	0	5017	0	2413	1
0	1	1	2	5601	1	8721	1	0	0	1	2	3141	1	5641	1
1	1	1	0	11501	0	4665	1	0	1	1	2	5133	1	7489	1
0	0	1	1	285	1	12621	1	0	0	1	1	1969	1	13289	1
0	0	0	3	425	1	11489	0	0	1	0	6	153	1	12957	0
0	0	0	4	13514	0	149	0	1	1	0	2	1193	1	11921	0
0	0	0	3	105	1	15417	0	0	1	0	3	1121	1	11273	0
1	1	0	1	173	1	16533	0	0	1	0	4	177	1	12877	0
0	0	0	3	12207	0	89	0	0	1	0	2	69	1	12493	0
0	0	1	3	337	1	8108	1	0	0	1	5	6597	0	89	0
1	1	1	4	6101	0	105	1	1	1	1	2	5949	0	145	1
0	1	1	5	5925	0	141	1	0	0	1	2	6393	0	97	0
0	1	1	4	5641	0	69	0	0	1	1	3	8193	0	97	0
0	0	1	3	7933	0	177	1	0	1	1	2	73	0	93	0
0	0	1	2	3297	1	3897	1	0	0	1	2	4977	0	149	1
0	1	1	2	149	1	4901	1	0	0	1	1	8357	0	157	1
0	0	1	3	3197	1	1893	1	0	0	1	4	2993	1	1765	1
0	0	1	4	81	0	85	0	0	1	1	2	1689	0	105	1
0	0	1	2	2417	1	1921	1	1	0	1	2	1557	0	233	1
0	0	1	1	1205	1	3065	1	0	0	1	5	893	1	1677	1
1	1	1	2	2789	0	189	1	0	0	1	4	581	1	2845	1
0	1	1	2	2609	0	105	1	0	0	1	3	1157	1	2417	1
0	0	1	4	685	1	2733	1	0	0	1	4	1010	1	2753	1
0	0	1	1	3021	0	109	1	0	0	1	3	2077	0	633	1
0	0	1	4	1041	1	5349	1	0	0	1	6	165	1	4296	1
0	1	1	4	761	1	5457	1	0	0	1	3	1173	1	3812	1
0	1	1	3	321	1	4385	1	0	0	0	6	730	1	3534	1
0	1	1	2	5449	0	201	1	0	0	N/A	N/A	N/A	N/A	N/A	N/A
0	1	1	3	69	1	3677	1	0	0	N/A	N/A	N/A	N/A	N/A	N/A
0	0	1	2	2249	1	3737	1	0	0	0	2	3657	0	2041	0
0	1	1	1	2469	1	5941	1	0	0	0	2	89	0	125	0
0	0	1	1	4693	0	749	1	0	0	0	3	1893	1	4469	0
0	0	1	0	85	0	1629	1	0	1	0	2	1205	0	61	0
0	1	1	1	469	0	1993	1	0	1	0	3	409	0	165	0